



Guideline for Integrated Watershed Management Data Collection: National Standard



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Citation

Desta G; Adimassu Z; Yaekob T; Taye G; Woldearegay K; Tibebe D; Abera W; Erkossa T; Tamene L. (2025). Guideline for Integrated Watershed Data Collection: National Standard. Ethiopian Institute of Agricultural Research (EIAR), Addis Ababa, Ethiopia. 79 p.

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Design and layout: Diego Murillo
Cover photo credit: IWMI / Matthew McCartney

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April 2025

Acknowledgments

We extend our heartfelt gratitude to the Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ), the Bill & Melinda Gates Foundation (BMGF) through Supporting Soil Health Interventions in Ethiopia (SSHI), the Alliance of Bioversity International and CIAT, and Accelerating Impacts of CGIAR Climate Research for Africa (AICCRA). Their invaluable technical and financial support was instrumental in producing this guideline. Our sincere thanks go to the reviewers whose insightful inputs, comments, and suggestions played a vital role in shaping the content. We also acknowledge the Coalition of the Willing (CoW) and its task force for involving us in working on this important guideline and for their consistent support, encouragement, and collaboration. We would like to give special recognition to Prof. Mitiku Haile for his coordination role and motivation throughout the write-shop.

Foreword

I am delighted to present this national guideline for standardizing integrated watershed management (IWM) data collection methods developed through the collaboration of national and international researchers. This guideline represents a significant milestone in the advancing and critical evaluation of IWM practices. Its development was initiated by the Coalition of the Willing (CoW) and facilitated with support from EIAR, CIAT, and GIZ. The Ethiopian Institute of Agricultural Research extends its deepest gratitude to all contributors for their dedication and teamwork in bringing this initiative to fruition.

The EIAR acknowledges this guideline as a critical addition to a broader series of data standardization guidelines. These include guidelines on agricultural water management, soil microbiology, soil fertility, and soil surveys, which have already been published and adopted by various stakeholders. These initiatives collectively play a pivotal role in harmonizing data.

This IWM guideline focuses on key areas such as information/parameters to be collected, instructions for data collection, and measurement and analysis techniques covering watershed delineation, characterization, design, implementation, monitoring and evaluation, and impact assessment stages.

Building on prior initiatives, the IWM guideline stands as an exemplary model for other sectors in agricultural research. It demonstrates a well-structured approach to data standardization and underscores the value of collaboration among experts from diverse fields. This collective effort paves the way for creating an integrated system for generating, storing, and sharing agricultural data efficiently and effectively.

The EIAR is dedicated to incorporating this guideline into its operations and advocating for its adoption across the national agricultural research system. On behalf of the EIAR, I express sincere gratitude to all contributors and reviewers who have invested their time and expertise in this initiative. Your invaluable contributions have been instrumental in creating a resource that will greatly enhance our capacity for data standardization and use. We firmly believe that this guideline will become an essential tool for researchers, practitioners, and policymakers, enabling them to unlock the transformative potential of data in integrated watershed management.

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Preface

Integrated watershed management (IWM) is a holistic approach to sustainably managing land, water, and ecosystems within a watershed. It is an essential strategy for addressing challenges such as land degradation, biodiversity loss, water scarcity, and climate change while ensuring the livelihoods and well-being of communities. In Ethiopia, where the agricultural sector is central to the economy and rural livelihoods, effective watershed management is critical for sustainable development.

This Integrated Watershed Management Guideline has been developed as a comprehensive resource to standardize data collection, analysis, and management practices across the country. By establishing uniform methodologies, this guideline aims to enhance the consistency and reliability of watershed data, thus fostering better communication, planning, decision-making, and implementation of IWM strategies.

The creation of this guideline reflects the multi-disciplinary collaborative efforts of national and international experts, practitioners, and development partners. Facilitated by the Ethiopian Institute of Agricultural Research, CIAT, and GIZ, this initiative underscores the importance of partnership in addressing complex watershed management challenges. It builds on existing frameworks and incorporates global best practices tailored to Ethiopia's unique environmental and socioeconomic contexts.

This guideline is designed to support a wide range of stakeholders: researchers, policymakers, development practitioners, and local communities. It provides practical tools and methodologies for designing, collecting, and using data on hydrology, soil, land use, and socioeconomic aspects within watersheds. By doing so, it seeks to empower stakeholders to implement integrated nature-based solutions that balance ecological integrity with human needs.

We firmly believe that this guideline will play a pivotal role in advancing Ethiopia's efforts to achieve sustainable watershed management. It aligns with national priorities for environmental conservation, climate resilience, and agricultural productivity, contributing to the broader goals of food security and rural development nationally while addressing the United Nations' Sustainable Development Goals.

On behalf of the EIAR and all contributing partners, we extend our deepest appreciation to everyone who contributed to this significant work. Your expertise and dedication have shaped a resource that will serve as a foundation for integrated watershed management in Ethiopia for years to come.

Finally, we invite all stakeholders to embrace and use this guideline as a valuable tool for fostering sustainable and inclusive development across the nation's watersheds.

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Glossary of Terms

Baseflow	River flow without surface runoff.
Bathymetry	The measurement of the depths of lakes, reservoirs, dams, and other large bodies of water.
Data	Information in raw or unorganized form such as measurements or statistics used as a basis for reasoning, discussion, or calculation.
Experimental plot	A plot used to evaluate different soil conservation practices and land use systems.
Geomorphology	The study of landforms, their processes, form, and sediments at the surface of the Earth.
Gully erosion	Form of erosion consisting of open, unstable channels that have been cut more than 0.3 m deep into the ground.
Landslide	Movement of a mass of rock, debris, or earth down a slope.
Local knowledge	A collection of facts and information related to the entire system of concepts, beliefs, and perceptions that people hold about the locality around them. This includes the way people observe and measure their surroundings, how they solve problems, and how they validate new information.
Mapping unit	The smallest mapping area where there are homogeneous land features such as slope, soil, land cover, geomorphology, etc.
Metadata	Data that provide information about other data; in short, data about data.
Pluviometer	Equipment used to record rainfall continuously on a recording chart.
Rill erosion	Small channel of linear erosion resulting from concentrated overland flow.
Runoff	An overland flow of excess rainfall.

Runoff plot	A bounded or unbounded plot used to measure runoff and soil loss for a given area over a given time.
Sediment storage capacity	The capacity of the bund basin area to retain and store washed sediment.
Sediment yield	The amount of sediment transported at a watershed outlet in a given time period.
Inter-rill erosion	Erosion resulting from a uniform removal of topsoil by overland flow.
Splash erosion	Erosion resulting from detachment of soil by raindrop or runoff impact.
Stream discharge	A total river flow – the sum of runoff and baseflow.
Sub-watershed	Smallest hydrological unit of a watershed that drains into a specific location such as the tributary of a river.
Sustainable land management	Practices and technologies that aim to integrate the management of land, water, biodiversity, and other environmental resources to meet human needs while ensuring the long-term sustainability of ecosystem services and livelihoods.
Tillage erosion	Displacement of soil downslope upon tillage operation.
Toposequence	Land configuration along the slope.
Water management	Management of surface and subsurface water resources for societal and environmental needs.
Watershed	A hydrological land surface unit draining water to a single outlet.
Wind erosion	Movement of soil particles and organic materials from one place to another by the action of wind energy.

Acronyms

Unless defined, the following are acronyms for the following words in the text:

ACED	Assessment of Current Erosion Damage
AP	Agroforestry practices
AT	Agronomic techniques
CoW	Coalition of the Willing to enable soil and agronomy data sharing and access
DEM	Digital Elevation Model
FAO	Food and Agriculture Organization of the United Nations
GIS	Geographic information systems
GLASOD	Global Assessment of Soil Degradation
GLCF	Global Land Cover Facility
GPS	Global positioning system
IWM	Integrated watershed management
LADA	Land Degradation Assessment for Dryland Areas
LM	Land management
LMP	Land management practices
LULC	Land use and land cover
MM	Mechanical measures
MoARD	Ministry of Agriculture and Rural Development
MU	Mapping unit
RS	Remote sensing
RT	Revegetation techniques
SCRIP	Soil Conservation Research Programme
SI	International System of Units
SLM	Sustainable land management
SMT	Soil moisture techniques
SWC	Soil and water conservation
UR	Use right
VM	Vegetative measures
WLRC	Water and Land Resource Center
WMO	World Meteorological Organization



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1. Introduction

With a population surpassing 100 million, Ethiopia is a country whose economic growth heavily depends on agriculture (CSA, 2016). The country has diverse topographical, geological, geomorphological, and hydrological conditions – creating diverse agroecological zones where the central, southeastern, and northern highlands are hilly/mountainous draining to lowlands. Because of the rapid increase in population, most of the agricultural land is occupied and limited land is left for agricultural production in the highlands, except through integrated hillside/mountain development. Smallholdings for agriculture are highly fragmented and degraded, leading to low agricultural productivity and associated water scarcity and food insecurity. The natural ecosystems (e.g., forest, shrub, grazing land/rangeland) are alarmingly converted to agricultural production even on marginal hillslopes. As a result, our perennial streams, reservoirs, and drinking water supplies are being degraded and threatened.

Integrated watershed management (IWM) is a key approach to sustaining agricultural economic development and maintaining natural resources. We need to protect and restore environmental quality in our agricultural production systems, river basins, and other land uses. The government recognizes that the complex interactions of natural and human systems (continuous crop cultivation, deforestation, steep slope cultivation, unregulated water withdrawals, erosion and sedimentation, competition for land uses, non-point source pollution, etc.) coupled with unregulated/unplanned expansion of human settlements into fragile ecosystems have been creating new management problems that cannot be addressed by business-as-usual practices. Human modifications of land and water directly affect the delivery of water, sediments, and nutrients, and thus fundamentally alter terrestrial and aquatic ecosystems. Such unregulated and unplanned management is also contributing to hazards and disasters such as mass movements, flooding, extreme drought, and associated problems. Addressing all the problems that concern land, water, vegetation, and other ecosystem resources of a watershed in any way comes under watershed management strategies. The government therefore developed the Ethiopian Strategic Investment Framework (ESIF) for sustainable land management (SLM) and determined that a comprehensive watershed management process would be the most efficient means to address these problems (MoARD, 2010).

Watershed management is an iterative process of integrated decision-making regarding uses and modifications of land and water within a watershed (Joseph, 2009). This process provides a chance for stakeholders to balance diverse goals and uses for environmental resources, and to consider how their cumulative actions might affect the long-term sustainability of these resources. This process requires research supported by standardized scientific methods/tools that generate evidence on the impacts of watershed interventions. Data should be collected at different scales and documented with standard methods and procedures. This would enable problem-solving research, promote evidence-based decisions, address common watershed-related constraints, and support better communication among researchers.

Research data can include characterization of watersheds as well as monitoring and evaluation of interventions. Although Ethiopia has been implementing extensive watershed management for the last five decades, standardized data collection methods and evidence documentation are lacking. The data for evaluating the performance and multi-dimensional impacts of watershed management interventions are not adequate for making evidence-based decisions. This calls for reviewing and standardizing data generation processes for IWM research and development endeavors. Thus, this standard for data collection encompasses data collected in plot, farm, sub-watershed, and watershed studies.



2. Purpose of the Guideline

Integrated watershed management is an approach to nurture the integrated use and management of land, vegetation, water, and other watershed resources to benefit people, protect the watershed ecosystem, and reduce or avoid negative downstream impacts (German et al., 2007). It is meant to manage human activities and natural resources together in a watershed as a system. Changes in environmental costs and socioeconomic impacts result from natural soil erosion, changes in farming systems, over-abstraction of water, overgrazing, deforestation, and pollution. Watershed management encompasses interventions that meet diverse stakeholder and ecosystem needs under various land uses and farming systems: highland agriculture farming system, lowland agriculture farming system, forest system, agro-pastoralist and pastoralist system, and upstream and downstream communities. Depending on the prevailing conditions, watershed management could have one or more objectives: agricultural production and food security, water management and security, drought management, flood management, conservation of biodiversity, and so on. Nevertheless, because of the strong interconnection among watershed resources, watershed management interventions focusing on a single component of the watershed or single objective might result in multiple and far-reaching impacts.

We recognize that watersheds constitute both human and natural systems. However, *this guideline focuses mainly on natural systems and processes (soil, water, vegetation, climate, and topography interaction) and is limited to current socioeconomic and livelihood-related interventions. This guideline adopts a micro-watershed scale of a minimum size of 500 ha, which is aligned with the national community-based integrated watershed management guideline.*

We used the following assumptions in preparing this guideline:

- ▶ Many watersheds do not constitute a full-fledged data monitoring facility and have field experiments to employ advanced data collection methods.
- ▶ Selected methods and procedures presented in this guideline are identified based on the technical capability of most experts and average watershed conditions.
- ▶ The type of data requirements in a watershed are considered based on systems thinking.
- ▶ Interaction of local knowledge and empirical testing methods is taken into consideration.
- ▶ Future updates are necessary to reflect advanced methods and tools for watershed monitoring and evaluation (such as automatic data loggers, radar, photogrammetry, remote sensing, etc.) as well as the dynamics in watershed management. Regular updates ensure the workability of the data collection standards in relation to inclusiveness and utility by most experts and researchers.

The main purpose of standardized data collection methods for integrated watershed management is *to establish minimum common standard procedures and methods of data collection (sampling, mapping, describing, and measuring) for a range of research, impact evaluation, and monitoring work commonly*

conducted in a watershed or at watershed scale. The guideline emphasizes the importance of (a) a systems approach for watershed analysis and management, that is, understanding how the watershed functions, and (b) using appropriate but widely used or applied procedures and methods that can easily be adapted by most researchers and expert groups. The methods also attempt to incorporate the importance and interaction of local knowledge and empirical testing methods to build co-production of knowledge, technology, innovation, and management practices.

A photograph showing two men in a rural setting. One man, wearing a brown long-sleeved shirt and brown pants, is crouching and operating a yellow manual pump. The other man, wearing a grey and white jacket, is also crouching and looking at the pump. A green pipe is connected to the pump and extends towards a body of water. The background shows a pond with some reeds and a grassy bank. The text '3. Important Considerations' is overlaid on the image in a large, bold, white font with a black outline.

3. Important Considerations

© IWM / Maheder Hallekiasse

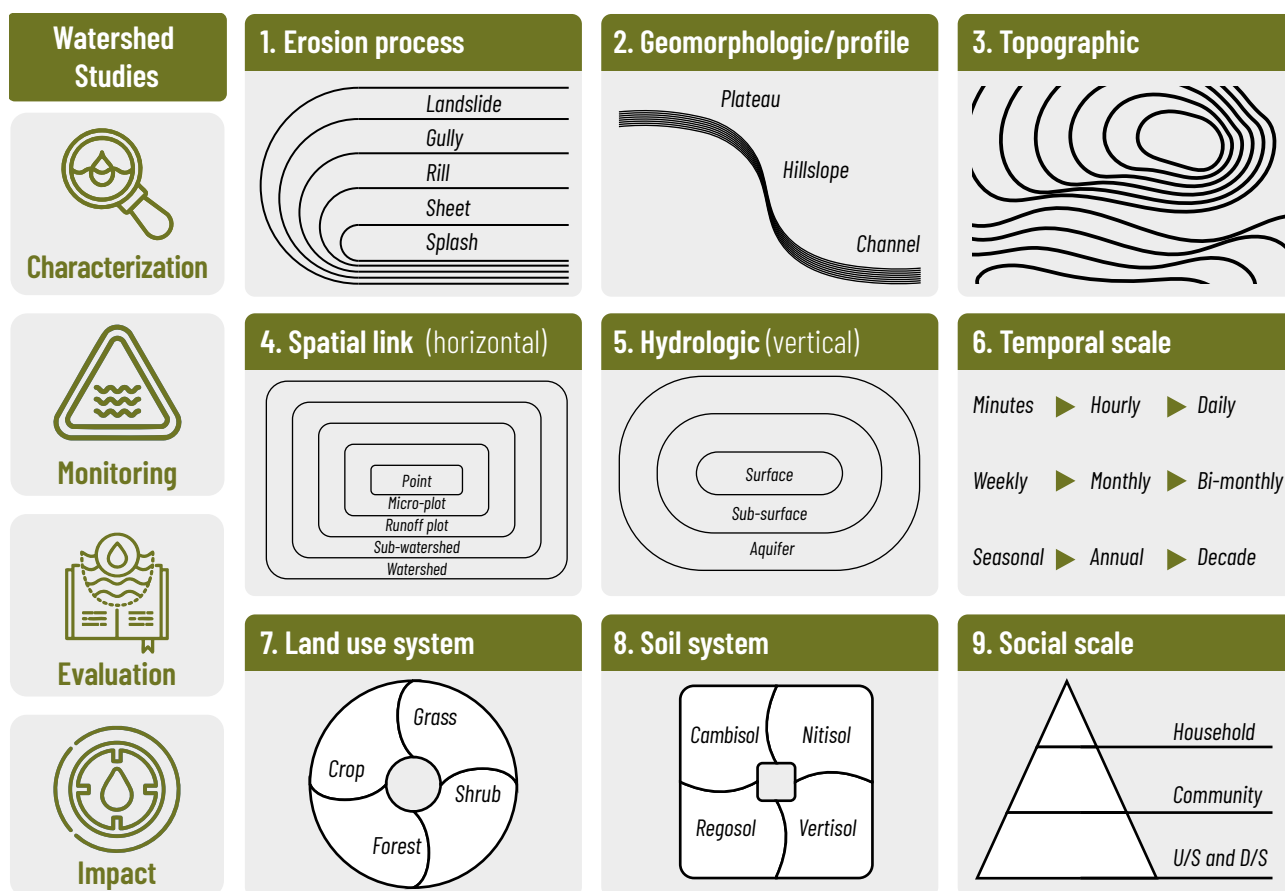
This standardization of data on integrated watershed management is guided by the concept and principles of watershed management. The identification of data types and associated methods follows the principles of watershed processes and interactions as well as accounting for the data/information demand from diverse stakeholder perspectives and needs. As illustrated in Figure 1, the data requirement and related methods are developed by taking into consideration the interactions within a watershed, which include

- i. Processes and interactions: topographic, hydrologic (surface and subsurface processes), geomorphologic, geologic, soils, climatic variables, land use and land cover, anthropogenic factors, and upstream-downstream issues.
- ii. Farming systems: farming system components and their spatial and temporal interactions.
- iii. Scales: spatial scale (plot to watershed), temporal scale (event, daily, monthly, seasonal, and annual), and social scale (household, community, and upstream/downstream).
- iv. Thematic and multidisciplinary aspects: geology, hydrology, geomorphology, soil, water, forest, and crop.
- v. Processes and interactions of natural and human factors.
- vi. Major watershed-related studies and monitoring: watershed delineation, characterization (baseline), monitoring, evaluation, and impact.

The following elements have been considered in developing this guideline for data standardization to be generated/collected through IWM projects and programs:

- **Scientific requirements (scientific, technical, and practical know-how):** Watershed management solutions demand harmonized and synergistic interventions to respond to complex processes and interactions within watersheds. Watershed data collection should be based on sufficient technical and systemic knowledge and sound data management skills. It requires a strong scientific base of interdisciplinary, transdisciplinary, analytical, and systems knowledge, which are produced with the integration of local knowledge.
- **Base maps:** Topographic base maps of watersheds/micro-watersheds with appropriate scale, which is adequate for a detailed survey within a watershed (*area from 500 to 1,000 ha*), are essential as they are the basis for benchmarking and the sampling, monitoring, and surveying of various data types and variables (land use, crop production, performance of soil conservation, assessment of erosion damage, geology, soil, etc.). *To conduct different research and monitoring at different spatial units, precise topographic base maps at least at a scale of 1:5,000 are appropriate.*

Figure 1. Illustration of the elements of heterogeneity, continuity, and interactions (soil, water, vegetation, upstream, and downstream) in a watershed approach.



Source: Authors' construction.

- **Uniform scale of data collection:** Thematic maps in a watershed should have the same scale to facilitate the transfer of information or the production of subsequent maps by superimposing one on another (conducting overlay analysis). *This can be achieved by preparing uniform data collection units (25 m × 25 m) for many attributes with the aid of GIS and GPS tools.* This offers the possibility to analyze parameters spatially and compare the different parameters under various prevailing management conditions.
- **Hierarchy of data collection:** To understand the relationship of different processes that are affected by the spatial aspects of soil, topographic, geologic, geomorphologic, hydrologic, land use and land cover, land management, and farming system elements and to achieve data transferability, *we could use a different hierarchy with different spatial scales (point, micro-plot, plot, household, community, sub-watershed, and/or watershed).* The research concept involves, among other things, several field- and watershed-level research experiments for comparative studies. For example, the functionality and degree of performance of interventions on gullies, water harvesting/water points, and other infrastructure can be surveyed and assessed at feature scales. Soil erosion processes and crop productivity data should be measured at micro-plot, plot, and field levels. To determine the spatial variability of the impact of land management practices on sediment yield and groundwater recharge, small stream flows and soil accumulation capacity of structures can be evaluated at sub-catchment scale. Climate, land use/land cover, land management distribution, geomorphology, geology, hydrogeology, discharge, erosion damage (rill and gully), and sediment yield are documented at watershed scale. Socioeconomic parameters (perceptions, adoption, and cost-benefit analysis) are measured at household and community scale.

- ▶ **Integrating scientific and local knowledge:** In this transdisciplinary approach, indigenous practices are considered in the research work and local stakeholders are engaged/involved. Pre-existing technologies and local indigenous knowledge should be used as an entry point to integrated watershed management. This helps to articulate and convert implicit knowledge into explicit knowledge. This could be done at all levels of IWM: during inception/planning, design/technology selection, implementation, and monitoring and impact evaluations.
- ▶ **Data interpretation:** All data should be carefully interpreted, because each method has its specific range of applications. *An application of data beyond their specific range should be avoided, as this might lead to unreliable information, thus misleading stakeholders and decision-makers in various aspects, including prioritizing interventions and finding appropriate nature-based solutions.*
- ▶ **Selection of methods:** The methods described in this guideline are assumed to be widely known and easily understandable and usable by experts from different disciplines with average technical knowledge in the subject areas. For example, the geological and hydrogeological parameters are determined by experts in the field. In this guideline, only limited methods and approaches are presented because many users have already been working in poor research facilities and this challenge is expected in using this guideline. Alternative methods are presented whenever relevant to provide more options for users depending on their working conditions (technical and research facilities). More advanced and sophisticated methods and approaches are not adequately presented here as the main aim of this guideline is to develop standardized data generation methods that could be used by most of the research and development personnel in the country. This ultimately provides an opportunity to build a national dataset on the subject matters that is adequate to inform evidence-based decision-making and promote better watershed management practices.



4. Approach

This guideline was developed by engaging national experts from universities, national agricultural research institutes, and CGIAR centers who have accumulated vast experience in research and development in watershed management. The data collection methods presented in this guideline were based on a review of various documents (reports, SLM guidelines, and published materials) and practical field research experiences of the experts involved in developing the guideline.

5. Standards of Data Acquisition

This section describes the standards of data acquisition: (a) unit of data sampling (data collection unit or scale), (b) data acquisition methods and procedures (for both primary and derived parameters), (c) unit of measurement of variables, (d) formats to be used for data collection, and (e) data quality control procedures. Data collection templates were developed to use uniform formats and simplify database development. Readme files and metadata were developed to provide a clear catalogue of the data.

5.1 Types of data requirements in watersheds

The types of data needed for delineating, characterizing, planning, monitoring, and evaluating watershed management interventions depend on watershed problems and their interactions, the size and heterogeneity of the watersheds, and management objectives. The data requirements constitute general data and thematic physical and socioeconomic data. The following data types describe broad aspects of required data. The detailed variables and parameters are presented in the subsequent sections under the different types of studies.

General data: Data on a watershed should include watershed name, location, administrative districts, boundaries, size/area, elevation, presence of streams, tributaries, agroecologies, farming systems, population, accessibility, roads, etc. The watershed might need to be divided into many sub-watersheds depending on watershed heterogeneity and each should be assigned a number for easy identification. Further data will be needed on a sub-watershed basis.

Basic physical data: Data on geology, soils, hydrology, and infrastructure can usually be obtained from existing reports or assessed using a field survey. Geomorphological data such as drainage patterns, stream density and order, channel profiles, landforms, slopes, aspect, etc., can be obtained and/or quantified by using a good topographic map to determine hydrological conditions and land capabilities of a watershed and appropriate conservation needs. Data on soil hydrologic conditions are required in different land use classes such as forest, grassland, rangeland, and cultivated land.

Constraints and community perception data: In watersheds, the interaction of human and natural systems is often complex. The community has diverse interests and needs, which involve conflicts of interest. This complexity should be characterized in terms of prioritized constraints through community participatory approaches. Community perception on watershed management interventions and overall impacts is needed to meet community needs. Upstream and downstream relations need to be understood to build benefit-sharing mechanisms.

Land degradation data: Data on the forms, distribution, and extent of land degradation are essential to characterize the severity of land uses and prioritize areas of the watershed for management interventions. These data are also useful to determine relationships with trends of production, ecosystem services, and food insecurity.

Climate, hydrology, and water resources data: Climatic data such as precipitation, wind, evaporation, temperature, humidity, etc., can normally be obtained from climatic stations in the watershed or nearby stations. For watershed management, especially for runoff estimation and design of erosion control measures, rainfall data (such as direction, amount, and intensity) are required. In this case, some automatic rain gauges might need to be placed in the watershed area distributed based on rainfall variations and rainfall directions in the whole watershed. Soil moisture dynamics and infiltration characteristics of watersheds are needed to understand soil and climate relations and the resulting hydrology of the watershed. Data on surface runoff, stream flows, and sedimentation are often not available in upstream watersheds and there is a need to collect data from fields and stations at sub-catchments and outlets of streams. Data on hydrogeology and groundwater levels should be evaluated and are often associated with variations in rock types (geological formation), geomorphology, and topographic conditions. The water balance or water budget of the watershed should be estimated. Water use problems regarding quality or quantity also need to be determined. For information on flood damage, landslides, drought, and other hydrologic problems, the usual techniques are field documentation of damaged areas, interviewing people affected by such hazards, and discussing the matter with knowledgeable institutions in the area.

Land use, land capability, and biological data: This category of data usually includes current land use, land use history and future trends, land capability or suitability, and several vegetation surveys. For current land use, a new survey is often required to identify forest and range cover, cropland, plantations, rural settlements, infrastructure, urban areas, water bodies, etc. The needed data and survey criteria for each major and detailed land use will depend on management objectives and input data requirements for modeling purposes. Land use history is needed to reveal past lessons and experience and evaluate the impacts of historical land uses on its status. These kinds of data can be obtained from reports, records, or knowledgeable persons in the local community.

Future trends in land use are very important to planners and decision-makers. Land capability or suitability data are usually required to show the limit, extent, and proper use of each piece of land in the watershed. Although criteria might vary, they are essentially based on soil, climate, slope, and land capability classification, which have been used in many hilly watersheds. Special biological, vegetation, and multiple-use surveys are sometimes needed. They might cover forest resources, protection forest, rangeland and grassland, riverine, and wildlife reservation areas.

Erosion data: Since erosion is a major concern in most watersheds, the collection of erosion data is a very important part of the overall survey. The causes of erosion and its severity should first be identified. Field-level runoff information and associated data on soil loss due to different forms of erosion (splash, sheet, rill, gully) are required to determine the spatial and temporal processes that affect erosion in the different parts of the watershed. Appropriate measurement and survey techniques should be used regarding the collection of data on erosion (splash, sheet, rill, gully, tillage, stream, and wind) as well as landslides and associated problems. Surveys of geology and landforms or geomorphology, both of which are related to erosion, are needed.

Data on performance of land management practices: Individual technologies and practices should be evaluated in terms of effectiveness and adoption to identify the most effective and adaptable options. Specific practices or technologies, as well as overall implementation, should be evaluated in terms of the impact on ecosystem services and community well-being.

5.2 Data standard for watershed characterization

Watersheds need to be characterized and documented at the start (baseline phase) before interventions in terms of constraints and community needs, soil and hydrogeology, geomorphology, infrastructure, and land use systems as a benchmark to evaluate watershed performance after implementing interventions.

5.2.1 Data on constraint analysis of watersheds

Watershed issues and constraints can be identified and prioritized by means of interviews, focus group discussions (gender and wealth rank disaggregated), and field observations through transect walks. At each interval of the transect survey (every 75 to 100 m along the transect depending on the heterogeneity of the watershed attributes such as land use, soil, vegetation, etc.), all biophysical and socioeconomic information must be recorded. Respondents must represent different community categories (such as age groups, from different villages, wealth status, and gender). The participants identify constraints of a social, economic, and environmental nature from broader dimensions, along with technical, legal, governance, and financial aspects. Discuss how to prioritize and screen the constraints to come up with a manageable number of constraints for pairwise ranking (Tables 1, 2, and 3).

Table 1. Ranking of constraints of a watershed using pairwise ranking matrix.

Code		1	2	3	4	...n	Results	
	Problems	C1	C2	C3			Score	Rank
1	Constraint 1 (C1)	X						
2	Constraint 2 (C2)		X					
3	Constraint 3 (C3)			X				
4	Constraint 4 (C4)				X			
...	n							

Table 2. Wealth ranking criteria used by the watershed community.

Criteria and (unit)	Rich (high income)	Middle	Poor (low income)
Total landholding (ha)			
• Cultivated landholding (ha)			
• Cash-crop holding (ha)			
• Woodlot holding (ha)			
Number of cattle			
Other sources of wealth/income			

Table 3. Matrix presenting (baseline) information from transect walk in the study watershed.

Features	Transect stop 1	Transect stop 2	Transect stop 3	Transect stop 4	Transect stop 5	Transect stop n**
Landform*						
Land use type (local name)						
Corresponding English name						
Relative importance (cropped area)						

Continues ►

Features	Transect stop 1	Transect stop 2	Transect stop 3	Transect stop 4	Transect stop 5	Transect stop n**
Farmer's main criteria						
• Soil type						
• Soil color						
• Available moisture						
• Soil texture						
• Vegetation						
Trees (number and name of trees)						
Farmer's management practices						
• Major crops and fertilizer						
Animals						
• Forage						
Farmer's conservation/agroforestry practices						
Water source						
Major constraint						
Potential improvements						

* It can be flat, shoulder, back-slope, hollow, spur, ridge, valley bottom, etc.

** The number of transect lines depends on the dynamics of the watershed in terms of landform, land use, soil types, vegetation cover, etc.

5.2.2 Infrastructure (roads, settlements, and social services)

Infrastructure (roads, settlements, etc.) development often affects runoff patterns of landscapes and could cause problems in landscapes, such as flooding, gully erosion, sediment delivery, and other cascading effects. If managed, runoff water from such infrastructure could be used for productive purposes. There is therefore a need to evaluate infrastructure-landscape interactions in watershed development and management. The approach/method to be followed includes (a) an inventory of existing infrastructure and services (roads, settlements, etc.); (b) evaluation of the effects/damages of runoff water from such infrastructure on the landscape (gullies, flooding, landslides, etc.) and vice versa; and (c) assessments of the possibilities for managing runoff water from such infrastructure for use for various purposes, dissipation to safe locations, and/or design of mitigation measures. Table 4 summarizes the data and key parameters to be documented and evaluated.

Table 4. Data needed and parameters to be assessed in order to evaluate roads and settlements in watershed management.

No.	Parameter	Description
1	Location	Name of place
		Latitude, longitude, elevation
2	Type of road	Highway
		Feeder
3	Type of settlement	Village, town, city
4	Population: distribution pattern and density (number of people/km ²)	
5	Access to basic services	Health, energy, education, water sources
6	Road damage with water from roads/settlements	Sedimentation: clogging of drainage systems, deposition of sediment over roads, nature of the deposited sediments
		Gully initiation, runoff flooding, and mud flows
		Scouring: road drains, road subgrade, embankment fill, subgrade, cut slope
7	Land damage with water from roads/settlements	Waterlogging: upstream, downstream
		Spring alteration: drying, new springs, increase in discharge
		Gullies: upstream, downstream
		Mass movements
		Land subsidence
		Sedimentation: upstream, downstream
		Concentrated runoff: upstream, downstream
		Pollution sources and types

Continues ➤

No.	Parameter	Description
8	Practice of runoff water management from roads/settlements	Floodwater spreading (uncontrolled, controlled, guided)
		Storage (soakways, earth dams, water tanks, subsurface dams, groundwater recharge)
9	Potential for positive use of water from roads/settlements	Artificial or natural pits
		Infiltration structures in gullies, check dams, cut slopes (drainage masks), embankments
		Areas prone to water spreading
		Drainage collection

Note: This information should be integrated with geology, hydrogeology, geomorphology, and land use/cover attribute maps.

5.2.3 Geology, geomorphology, and hydrogeology

Understanding the geological, hydrogeological, and geomorphological conditions of a watershed is important for the proper planning, design, implementation, and monitoring of watershed interventions.

Geological and hydrogeological parameters: The watershed should be characterized in terms of rock and soil type, permeability, infiltration capacity, aquifer characteristics, and erodibility. This requires evaluating the (a) type, thickness, distribution, and characteristics of the unconsolidated deposits/soils; (b) type, thickness, distribution, degree of weathering, and permeability of the different rocks; (c) distribution and characteristics (infill, aperture, continuity, spacing, etc.) of the major geological structures (such as lithological contacts, faults, shear zones, foliations, bedding planes, joints, unconformities, and other weak zones); (d) groundwater aquifer systems as well as the recharge-storage-discharge zones; (e) inventory of existing groundwater and spring points, along with their yield and quality; and (f) erodibility of the soils and rocks in the watershed.

Geomorphological parameters: The geomorphological parameters to be evaluated in the watershed are (a) terrain characteristics and drainage patterns, (b) mass movement hazards (landslides, major gully zones, piping/subsidence, etc.), and (c) flooding hazards.

Methods/approaches: The above-mentioned data could be gathered from geological and structural mapping, hydrogeological mapping and characterization, and subsurface investigations (test pits and trenches) (Table 5). The geological, hydrogeological, and geomorphological investigations shall include:

- ▶ Review of previous work, including available geological, hydrogeological, engineering geological/geotechnical, geomorphological, and other information in the watershed and in other areas with similar conditions.
- ▶ Interpretation of DEM (at least 30-m resolution): this would be to identify lithological contacts, major geological structures (faults, shear zones, and lineaments), and hydrogeological units, and to assess the overall geomorphological conditions of the watershed.
- ▶ Geological, hydrogeological, and geomorphological mapping: this involves descriptions of the lithological conditions of the soils/rocks and structural mapping (from surface exposures, river sections, and geophysical survey and borehole logs, if available).

- ▶ Inventory of groundwater and spring-water points.
- ▶ Measuring: (a) the level, quality, and yield of groundwater wells, and (b) discharge and quality of spring water.
- ▶ Test pit/trench excavations, in situ (field) testing of soils/rocks, including infiltration, and permeability tests using different methods such as inverse auger hole methods.
- ▶ Sampling of soils, rocks, and water for laboratory analysis to evaluate their properties in relation to the type of watershed intervention.
- ▶ Selection of appropriate sites and installation of groundwater-level monitoring devices.
- ▶ Assessment of suitability of the soils/rocks in the catchment as construction materials, mainly clay, sand, and gravel; quarry sites for stone and crushed aggregates.

Minimum mapping scale: 1:2,500 (25 m × 25 m).

Table 5. Data needed and parameters to be assessed during geological, hydrogeological, and geomorphological mapping.

No.	Parameter	Description
1	Location	Name of place
		Latitude, longitude, elevation
2	Rocks, soils, and major geological structures	Rocks: types, distributions, and hydraulic properties
		Soils: types, distributions, and hydraulic properties
		Major geological structures
3	Hydrogeological parameters	Aquifer properties and their distributions
		Groundwater levels
		Groundwater flow direction
		Groundwater quality (field level)
		Groundwater recharge
		Spring water (discharge, quality)
4	Geomorphological parameters	Slope gradient (%)
		Aspect
		Slope shape (concave, convex, planer)

Continues ▶

No.	Parameter	Description
		Drainage density (m/km ²)
		Mass movement/landslide (type, length, width, depth) and its effects
		Erosion (sheet, gully, piping)
		Flooding
		Other geomorphological processes
5	Existing groundwater recharge practices	In situ water harvesting
		Ex situ water harvesting
6	Watershed management impacts	Changes to water table (elevation)
		Changes in groundwater flow direction
		Changes in infiltration/recharge/discharge rates and volumes on varying time scales
		Changes in baseflow
		Impacts on water quality
		Impacts on nearby receiving surface waters
		Others, if any

Note: A map that shows the geological (lithological and structural) features, major hydrogeological units, and geomorphological units shall be produced.

5.2.4 Watershed morphology

- a.** Total stream channel length (sum of the lengths of all drainages in a watershed)(km), L_s .
- b.** Mainstream channel length or longest channel length (km), L_m .
- c.** Watershed length or straight-line length (the length of a link, Strahler stream, or any other channel segment as measured along a straight line that connects the two endpoints of the segment)(km), L_b .
- d.** Length centroid to outlet (length from center of catchment to outlet of watershed along-channel slope)(km), L_{ca} .
- e.** Mainstream channel slope (along-channel slope)(average slope of mainstream channel)(%), S_l .
- f.** Watershed slope or straight-line slope (average slope of watershed)(%), S_b .
- g.** Drainage density (ratio of the total length of all streams within the watershed to the watershed area) (km/km²), D_d .

- h.** Source density (ratio of the number of sources in a stream network to the area of the corresponding drainage watershed) (no./ha), D_s .
- i.** Watershed shape factor (the ratio of the square root of watershed area to watershed diameter), S_f .
- j.** Strahler number (highest no. of stream order).
- k.** Network magnitude (the number of channels in each river network or sub-network that have a Strahler stream order of 1) (no.), N_m .
- l.** Sinuosity index (a dimensionless quantity and a measure of meandering of the stream). It is the ratio of the distance measured between two points along a stream to a measure of straight-line distance between two points, S_i .
- m.** Bifurcation ratio (R_b) is used to express the ratio of the number of streams of any given order to the number of streams in the next higher order, R_b .
- n.** Form factor/shape (tick), (a) fan shaped; (b) fern shaped
Form factor, $F = \frac{A}{L^2}$, where A = area of watershed (m^2) and L = length of watershed (m).
- o.** Compactness coefficient (CC) = $\frac{P}{C}$, where P = perimeter of a watershed and C = circumference ($C = \frac{P}{2\pi r}$) of a circle with area equal to that of the watershed.
- p.** Stream density (D), $D = \frac{\sum_i^n L}{A}$, where L = total stream length (m) and A = area (m^2).
- q.** Stream frequency (S_f) = $\frac{S_n}{A}$, where S_n = total number of streams and A = area (m^2).
- r.** Mean elevation (E) = $\sum_i^n e / N$, where e = total sample elevation at grid square and N = total number of samples.
- s.** Mean slope (S) = $\frac{\sum_i^n s}{N}$, where s = total sample slope at grid square and N = total number of samples.
- t.** Relief (R) = $H_h - H_l(m)$, where H_h = highest elevation (m) and H_l = lowest elevation (m).
- u.** Relief ratio (RR) = $\frac{R}{L_b} (km/km)$, where R = relief (difference between the highest and the lowest elevation, km), and L_b = total length from the peak of the watershed to the lowest point (outlet).

5.3 Data standard for watershed monitoring

5.3.1 Climatic variables

Rainfall amount

Rainfall is measured as the amount of water that reaches the horizontal ground or the horizontal ground projection plane of the Earth's surface and is expressed as a vertical depth of water in mm.

Rain gauges are classified into recording and non-recording types. The latter include cylindrical and ordinary rain gauges, and the observer manually measures precipitation with these types. Some recording types such as siphon rain gauges have a built-in recorder, and the observer must physically visit the observation site to obtain data. Other types such as tipping bucket rain gauges have a recorder attached to them, and remote readings can be taken by setting a recorder at a site distant from the gauge itself to enable automatic observation.

Rainfall data can be collected with a pluviometer at 1–1.5 m above the ground, and with a rainfall inclinometer (Table 6). Using a pluviometer, rainfall is recorded on a monthly or bimonthly chart, and the instrument is checked daily at 0900 for the smoothness of the recording, the calibration, and/or running of the clock.

These instruments are supplemented by daily rain gauges at two or more locations within each research watershed to observe differences in the spatial distribution of rainfall.

Instrument: Rain gauge

Frequency: Daily for manual recording at 0900 and at 15-minute intervals for an automatic rainfall recorder.

Table 6. Data format for recording daily rainfall amount.

Date	Amount/depth (mm)	Remarks
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Rainfall direction

Rainfall direction can be determined using an inclinometer, which computes the average weighted rainfall direction of each storm. Thus, each storm is represented by a three-dimensional vector, representing the mean angle and direction of daily rainfall (SCRIP, 2000) (Table 7).

Instrument: Inclinometer

Frequency: For each storm

Table 7. Data sheet for recording daily rainfall direction.

Date	Storm (number)	Rainfall from north (mm)	Rainfall from south (mm)	Rainfall from east (mm)	Rainfall from west (mm)	Remarks
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Temperature

The heat balance difference in the atmosphere between regions creates temperature distribution. This temperature distribution generates wind currents along with cloud and rainfall phenomena. Thus, atmospheric temperature is one of the most important meteorological elements as well as wind and precipitation.

Temperature should be recorded daily by reading minimum and maximum thermometers at a height of 1.5 m from the ground surface (for air temperature, Stevenson screen) and 5 cm above the soil surface (for surface temperature, under shelter) (SCRIP, 2000) (Table 8). In addition to the thermometer on the Stevenson screen, temperature data can be recorded using an automatic weather station. It is best to set up the automatic recording at 15-min intervals to align with the National Meteorological Service Agency dataset.

Instrument: Thermometer

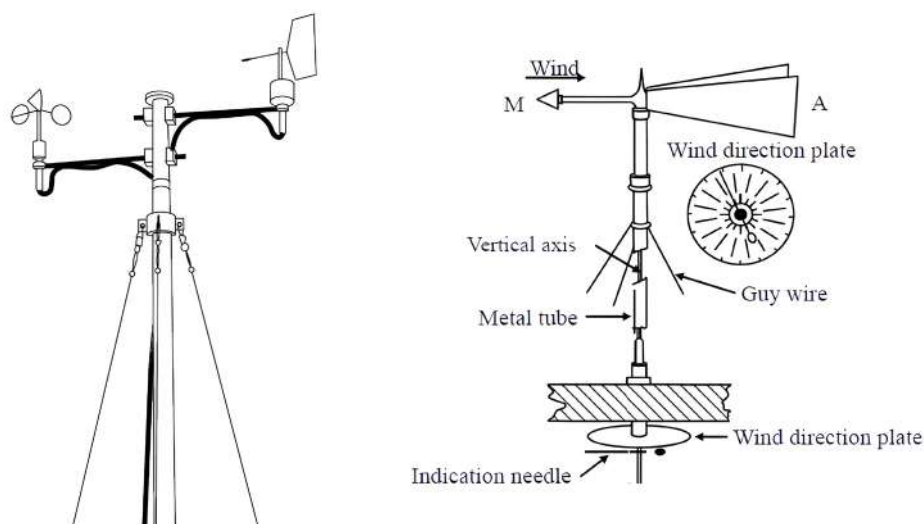
Frequency: Daily for Stevenson screen: minimum temperature at 0900 and maximum temperature at 1800 and 15-min intervals for an automatic weather station.

Table 8. Daily temperature recording sheet.

Date	Maximum air temperature (°C)	Minimum air temperature (°C)	Max. surface temperature (°C)	Min. surface temperature (°C)	Remarks
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Wind speed

Wind speed can be recorded using an anemometer, usually at 2 m above ground level (Figure 2). Several different units are used to indicate wind speed, such as meters per second (m/s), kilometers per hour (km/h), miles per hour (mph), feet per second (ft/s), and knots (kt). Later, data should be converted to standard SI units (preferably, km/h).

Figure 2. Anemometer for measuring wind speed and wind vane for measuring wind direction.

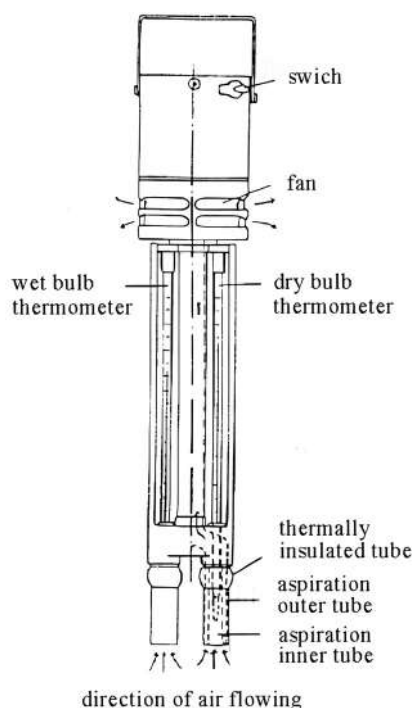
Source: Desjardins et al., 1986.

Wind direction

Wind direction is described in terms of the direction from which it blows. Wind direction can be recorded using a wind vane at least in terms of north, northeast, east, southeast, south, southwest, west, and northwest. If required, wind direction could be represented by one of the 16 directions.

Relative humidity

Relative humidity is the ratio of the current absolute humidity to the highest possible absolute humidity. A device to measure relative humidity is called a hygrometer. When water or ice covers the bulb of a thermometer (wet-bulb), latent heat is removed from the surface of the bulb as the water evaporates, and the wet-bulb temperature becomes lower than the air (dry-bulb) temperature. At a lower humidity, water evaporates more actively, so that the wet-bulb temperature lowers sharply. An aspirated psychrometer (Figure 3) measures humidity by measuring the difference between the dry-bulb temperature and wet-bulb temperature. Vapor and saturated vapor pressure tables (provided together with the instrument) should be used to calculate the relative humidity from the bulb reading (Table 9).

Figure 3. Structure of an Assuman-type aspirated psychrometer.

Source: Barber and Gu, 1989.

Instrument: Hygrometer

Frequency: Five times per day at 0600, 0900, 1200, 1500, and 1800, and at 15-min intervals for an automatic weather station.

Table 9. Dry- and wet-bulb temperature recording sheet.

Date	Time	Dry-bulb thermometer reading	Dry-bulb thermometer temperature	Wet-bulb thermometer reading	Wet-bulb thermometer temperature	Remarks
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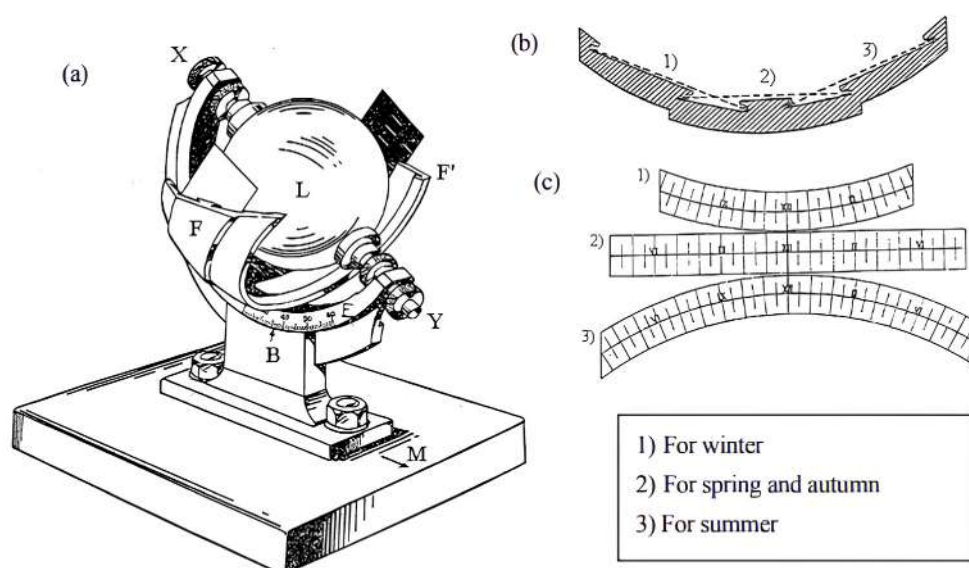
Sunshine hours

Sunshine duration is the length of time that the ground surface is irradiated by direct solar radiation (i.e., sunlight reaching the Earth's surface directly from the sun). In 2003, the World Meteorological Organization (WMO) defined sunshine duration as the period during which direct solar irradiance exceeds a threshold value of 120 W/m². This value is equivalent to the level of solar irradiance shortly after sunrise or shortly before sunset in cloud-free conditions. It is determined by comparing the sunshine duration recorded using a Campbell-Stokes sunshine recorder with the actual direct solar irradiance (Figure 4).

Solar radiation

Everything in nature emits electromagnetic energy and solar radiation. The energy of extraterrestrial solar radiation is distributed over a wide continuous spectrum ranging from ultraviolet to infrared rays. In this spectrum, solar radiation in short wavelengths (0.29 to 3.00 μm) accounts for about 97% of the total energy.

Figure 4. A Campbell–Stokes sunshine recorder for recording the duration of sunshine: (a) structure, (b) cross-section of bowl and grooves, and (c) recording cards.



Sources: Baumgartner et al., 2018.

Solar irradiance is expressed in W/m^2 and the total amount in J/m^2 . Conversion between the currently used unit (SI) and the former unit (calories) can be performed using the following formulae:

$$\text{Solar irradiance: } 1 \text{ kW/m}^2 = 1.433 \text{ cal/cm}^2/\text{min}$$

$$\text{Total amount of solar radiation: } 1 \text{ MJ/m}^2 = 23.89 \text{ cal/cm}^2$$

A radiometer is used to determine solar radiation. A radiometer absorbs solar radiation at its sensor, transforms it into heat, and measures the resulting amount of heat energy to ascertain the level of solar radiation. The radiometers used for ordinary observation are pyrheliometers and pyranometers that measure direct solar radiation and global solar radiation, respectively.

Table 10 shows the format for recording data from an automatic weather station.

Table 10. Data format for automatic weather station adopted from the National Meteorological Agency.

Date	Time	Relative humidity (% RH)	Wind speed (km/h)	Pyranometer SP-Lite and CMP3 (W/m^2)	Temperature ($^{\circ}\text{C}$)	Precipitation (mm)	Wind direction
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Evaporation

Evaporation can be measured with the help of a Piche tube evaporimeter (mL) at a height of 2 m from the ground surface (Stevenson screen) twice a day at 0900 and 1800. It can also be measured with the help of class-A evaporation pans (mm) at the ground surface. Alternatively, potential evaporation can be calculated using Penman–Monteith or Hargreaves equations.

Instrument: Piche evaporimeter and class-A evaporation pans

Frequency: Twice a day at 0900 and 1800 (for Piche evaporimeter)

Daily at 0900 and 1500 (for class-A evaporation pans)

5.3.2 Severity and extent of land degradation

Mapping the extent and severity of land degradation in a watershed helps to identify hotspots and priority areas of intervention. It is a basis for planning management options and helps ensure the rehabilitation and management of high-risk and more vulnerable areas. The general purpose is to provide standardized and objective data on land degradation; understand the nexus between land degradation, food security, biodiversity, climate, and overall ecosystem services; and create the basis for informed advice on land degradation at the local level.

Land degradation assessment is mostly done using Global Assessment of Soil Degradation (GLASOD) and Land Degradation Assessment for Dryland Areas (LADA) tools, which are developed by FAO. However, they are more appropriate for national and global assessments. For local land and water management planning, detailed assessment of land degradation locally is important. Thus, qualitative information on the severity and extent of land degradation can be collected using a homogeneous mapping unit of 25 m × 25 m (625 m²) for more heterogeneous watershed areas and 30 m × 30 m (900 m² or less) for relatively less heterogeneous watersheds (Kassawmar et al., 2018). The physically homogeneous mapping units are identified based on the homogeneity of combined features of land cover/uses, slope classes, soil types, landforms, geological formation, and other natural and man-made features. The homogeneous mapping units can be delineated and prepared in the office using high-resolution Google Earth images printed on A2 or A3 paper for field use. The delineation of mapping units is further refined (merged or split) and improved through ground truthing using GPS. Desta et al. (2019) describe the assessment of the degree/severity of land degradation in terms of soil depth, vegetation cover, erosion indicators/features (sheet, rill, gully, landslide), compaction, acidity, fertility status, agricultural suitability, and biotic functions. Estimates of the extent and severity of land degradation for each physically homogeneous mapping unit can be made using a transect walk by a team of experts and local people. For each unit, they discuss and describe the mapping unit in terms of the predefined criteria and decide the four severity classes (low, moderate, high, and very high). Table 11 presents a detailed description and the severity classes.

Table 11. Description of land degradation levels.

Degree of degradation	Description (of criteria)	Degree of degradation	Description (of criteria)
Low	Low erosion features (sheet) Moderate soil depth Moderate soil fertility status Restoration to full production is possible with management Original biotic functions are intact	High	Highly erodible and formation of rills and active network of gullies Extreme compaction/clay pan formation Very shallow soil depth (<10 cm) and gravel cover Less fertile/no fertilizer response Moderate acidity (pH of 4.0–5.5) Landscape is non-reclaimable at farm level Major engineering work required for terrain restoration Original biotic functions are largely destroyed/lost

Continues ➤

Degree of degradation	Description (of criteria)	Degree of degradation	Description (of criteria)
Moderate	Highly erodible with high inter-rill erosion and slight formation of rills Extreme compaction Shallow soil depth (<30 cm) Slight soil fertility status Agricultural productivity largely reduced but can still be used in local farming system Visible and apparent, including stunted crops or sparsely vegetated rangeland Major improvements are required to restore terrain to its productivity Original biotic functions are partially destroyed	Very high	Barren field with no vegetation Outcrops Extremely acid (pH <4) Extreme termite damage Landslides The terrain is unreclaimable and beyond restoration Original biotic functions are fully destroyed

Table 12. Severity and extent of land degradation data collection sheet.

Mapping unit	Area of mapping unit (ha)	Degree of degradation (light, moderate, strong, extreme)	Description (of criteria)	% of mapping unit affected	% of watershed	Land uses	Cause	Impacts	Remarks
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Quality check: The quality of land degradation severity and extent should be checked and evaluated by different experts and complemented with local knowledge (Table 12). When possible, the degradation extent must be assessed against quantitative estimates and remote-sensing techniques.

Note: The information is qualitative and subjective as it is derived from expert judgment. This sometimes leads to visual exaggeration when judging degradation from large mapping units.

5.3.3 Land use and land cover

Land use/cover data are required for many applications, such as local land use planning, development of national agricultural strategies, analysis of crop/erosion/hydrology model input parameters (for example, C and P factors of erosion modeling), analysis of climate change impacts, monitoring of national targets for land restoration and land degradation neutrality, water resource development and management, natural hazards assessments and mitigations, and so on. Nationally, only a few land use/cover datasets exist, such as those produced by the Ethiopian Highlands Reclamation Study in the 1980s, the Woody Biomass Inventory and Strategic Planning Project in the 1990s, a land use and land cover (LULC) map developed by FAO in the 1980s, and recent datasets developed by the Water and Land Resource Center (WLRC) (Hurni et al., 2015). These datasets do not have enough details for use at the micro-watershed level. To derive a land cover dataset that accurately represents the heterogeneity of the Ethiopian landscape, contextual approaches are

required that consider the complex biophysical, geomorphological, geological, and socio-cultural settings so that local settings could be properly captured. To meet the requirements of the different applications and capture detailed local land uses, a detailed level of classification that reasonably identifies the features is important (Table 13). Based on this, and considering the applications of the dataset, a classification scheme involving 30 distinct land use/cover classes (Hurni et al., 2015) needs to be defined.

Table 13. Land use and land cover classes.

Forest land	Shrub and bush	Grassland	Crop land	Others
Natural forest	Dense shrub and bush	Pastureland	Intensively cultivated land	Bare land
Plantation forest	Open shrub and bush	Open grazing	Extensive cultivated land	Degraded hills
Church forest		Controlled grazing	Agroforestry	Wetland
Riverine forest		Degraded/overgrazed	Irrigated fields	Settlement
Dense woodland		Savanna grassland	Homestead plantations	Water body
Open woodland		Range land	Shifting cultivation	Exposed rocks
Afro-alpine			Large-scale investment	Roads
Exclosures			Perennial crops	Railway
National parks and wildlife reserves			Fallow	

Methods: To develop large-scale land use/cover information, Landsat images are not appropriate for highly fragmented and heterogeneous terrains. Rather, high-resolution Google Earth images that are available online can be used. Ortho-photos, which are under development by the Ministry of Agriculture with a resolution of 40 cm, can also be used. Using Google Earth images, detailed homogeneous mapping units (MUs) for each micro-watershed can be developed. Each MU is defined based on the homogeneity of the landscape's basic characteristics such as land use/cover, terrain condition (altitude, slope angle, slope shape, aspect), soil erosion features, and landforms. Considering the possibility of mappable features (spatial configuration of land features) in the watershed and the required level of detail of information, the convenient smallest size of MUs can be defined to be 25 m × 25 m (625 m², or approximate data collection scale of 1:2,500 to 1:10,000). This allows us to generate the maximum possible detailed information such as settlements, small plantation/woodlots, and patches of forest around churches. The first task after developing the MU is labeling each MU with detailed LULC classes (Table 14). This can be done in the office by employing object-oriented screen digitization on a higher-resolution Google Earth image. The digitized MUs are produced on **A2 or A3 paper or uploaded to a Notepad** for the field work. During the field work, remapping of the MUs is done using GPS by taking ground truth. Whenever the boundary of the mapping units does not match, remerging and/or splitting is done. In each MU, the major predefined land use/cover classes and their extent are identified and collected based on expert and local people's judgment through a transect walk. To detect change in land use/cover, this method can be used to reconstruct land use/cover of previous periods by consulting key informants and focus group discussions with local opinion leaders and experts. Alternatively, despite its low resolution for use at local scale, LULC monitoring can be done regularly every 5 years. This can be modified depending on the objective of the land use change information.

Table 14. Land use and land cover (LULC) data collection sheet.

Mapping unit	Area of mapping unit (ha)	Land use/cover type	% of mapping unit	% of watershed
1				
2				
3				

Quality check: To ensure integration of local knowledge and compatibility of identified land use/cover classes with the farming system, we suggest consulting the community or opinion leaders.

Note: When appropriate and needed, parcel mapping helps to generate information on cropping systems.

5.3.4 Mapping land management practices

Research on land management practices (LMP) requires a multi-level monitoring approach, including intra-terrace, plot, field, landscape positions, sub-watershed, and watershed scales. Results gained with these multi-level monitoring methods eventually need to be interpreted together to get an overall idea of the order of performance, distribution, and magnitude of soil loss reduction and sediment retained by individual LMP and integrated ones under different conditions.

Type and distribution of land management practices and use rights

Data on the coverage of LMP and effectiveness are found to be relevant for evaluating the appropriateness of practices and performance of watersheds after intervention (Table 15). Detailed homogeneous MUs of 500 m² (up to 1,000 m² for less heterogeneous watersheds) can be used to identify the type and distribution of LMP that are implemented in watersheds (Table 16). Each MU is defined based on the homogeneity of the landscape's basic characteristics such as slope, land use/cover, soil erosion features, landforms, and soil types. These factors are assumed to influence the placement and selection of LMP. The MUs are developed by applying screen digitization on high-resolution Google Earth images and then produced on A2 or A3 paper for fieldwork. During field surveys using GPS, attributes of LMP (type, count, extent of coverage of practices, quality), land use rights (use right, land fragmentation/number of parcels per MU), and other attributes can be directly collected based on expert judgment and local knowledge for each MU through a transect walk. To facilitate data collection on LMP, guiding checklists are used. This whole process shall involve consultation with local experts and the community.

Table 15. Description of different land management practices

Type of practices	Description
Mechanical erosion control measures (MM)	No = no MM, soil bund (SB), stone bund (STB), bench terrace (BT), hillside terrace (HT), fanya juu (FY), trench (TH), micro basin (MB), eye-brow (EB), tie-ridge (TR), zai pit (ZP), cut-off drain (CD), waterway (WW), check dam (CD), ridge & furrow (RF), broad bed furrow (BBF), traditional ditch (TD)
Quality of mechanical erosion control measures	They can be described in terms of quality of structures and sediment accumulation efficiency. 1 = very poor, 2 = poor, 3 = average, 4 = good, 5 = very good
Soil management technique (SMT)	No = no SMT, C = composting, FM = farmyard manuring, GM = green manuring, IF = inorganic fertilizer, VC = vermicompost, L = lime, F = fallow

Continues ➤

Type of practices	Description
Vegetative method (VM)	No = no VM, GS = grass strip, S = Sesbania, T = tree lucerne, PP = pigeon pea, G = gesho, AD = <i>Acacia decurrens</i> , O = others
Quality of VM	It can be described in terms of adaptation of biological techniques in an agroecological setup. 1 = very poor, 2 = poor, 3 = average, 4 = good, 5 = very good
Agronomic technique (AT)	CS = crop selection, EP = early planting, CR = crop rotation, IC = intercropping, SC = strip cropping, CC = contour cultivation, MSC = multi-story cropping, M = mulching, ZT = zero tillage, MT = minimum tillage, SC = shifting cultivation
Agroforestry practice (AP)	No = no AP, AC ₁ = alley cropping, SC ₂ = scattered trees, AS ₄ = agro-silviculture
Re-vegetation (RT)	No = no re-vegetation, AC = area closure, A = afforestation with indigenous trees, P = plantation, CG = control grazing, PG = paddock
Land use right (UR)	C = communal, P = private, GU = group use, O = others



Table 16. Type, distribution, and coverage of land management practices.

MUs	Area of MU (ha)	Mechanical measures (MM1)	Mechanical measures (MM2)	Quality of MM	Soil management (SMT)	Vegetative measure (VM1)	Vegetative measure (VM1)	Quality of VM	Agronomy (AT)	Agroforestry (AP)	Re-vegetation (RT)	Use right (UR)	No. of parcels
1													
% of MU													
2													
% of MU													
3													
% of MU													

Note: For each MU, record two or more major types of practices and their quality/status in the first row and the percentage of coverage of the individual practice out of the MU in the second row. Quality of practices should be judged by a group of experts through consensus. These data will derive information on the total coverage of individual practices and integration of different management practices in a watershed.

5.3.5 Surface runoff monitoring

At a watershed scale, surface runoff processes can be studied. The dominant runoff processes depend on local conditions related to the catchment, slope gradients, vegetation cover, and antecedent soil conditions such as moisture content and level of groundwater. Horton overland flow (surface runoff) occurs when rainfall intensity exceeds the rate of infiltration capacity of the soil. This process dominates watersheds in arid and semi-arid environments with higher rainfall, soils or rocks with low infiltration capacity (e.g., clay soils, shales, etc.), and poorly vegetated landscapes.

Saturation overland flow occurs in those parts of the river basin where the soil is saturated throughout its depth due to rapid infiltration and the groundwater table thus reaches the soil surface. In the case of saturation excess conditions, saturated soil absorbs less water and excess water runs off the surface. Saturation might be caused by local rainfall and infiltration, by subsurface water flow from higher grounds, or by a rise in groundwater level through recharge.

Measuring runoff

Measuring surface runoff through runoff plots can be expensive in terms of cost and time. Therefore, to be cost effective, the same runoff plots can be used

- ▶ To measure runoff for design purposes (water harvesting)
- ▶ To measure soil loss (conservation planning)
- ▶ To evaluate the effectiveness of LMP

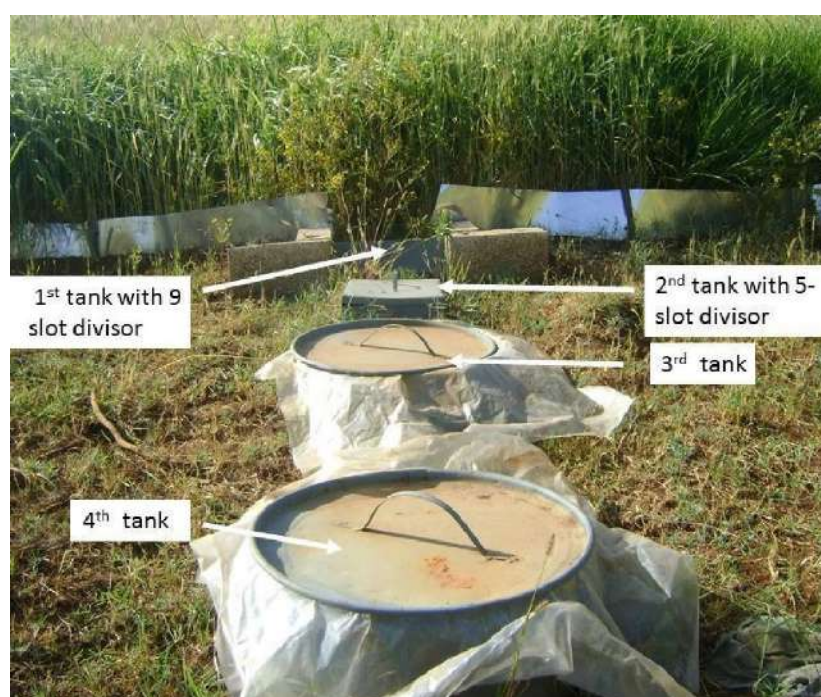
Runoff plots were commonly applied in many erosion and field hydrological studies. Long-term (several years) soil erosion studies in the U.S. by Wischmeier and Smith (1978) fixed 4-m-wide and 22-m-long plots as a standard at 9% slope. The SCRP in Ethiopia adapted 2-m-wide and 15-m-long runoff plots. Reviewing these against the topography, cultivation and tillage practices, and rainfall-runoff relations, we recommend runoff plot sizes of at least 4 m × 15 m for slopes <20% and 4 m × 10 m for slopes >20%. We use 4-m width rather than 2 m to capture the effect of off-contour tillage management on runoff concentration. Also, narrow runoff plots suffer from border effects with respect to data reliability. The suggestion of 10-m- or 15-m-long plots is concentrated in rill formation and development starting within 10 m or sometimes 5–6 m on steep slopes and within 15 m on gentle slopes (Gessesse et al., 2015; 2010). Runoff measurement follows the procedures listed here:

- a.** Conduct a survey to identify the major land use/land cover complex (including current land management), slope gradient, soil type, and geology.
- b.** With due consideration of the biophysical factors affecting runoff production (mentioned in “a”), locate a runoff plot for measuring surface runoff on representative land uses of the watershed.
- c.** Set up plots and install boundaries, runoff collecting tube/trough, runoff collecting tank, runoff and sediment sampling device, etc.
- d.** Measure the daily rainfall depth (mm) from the nearby rain gauge, runoff depth in the collector tank to determine area runoff depth (mm), and sample runoff + sediment to measure soil erosion rates (Table 17).
- e.** Measure runoff depth at three points in each collector to determine the average depth of runoff.

Table 17. Data format to collect rainfall runoff depths and sediment concentration from runoff plots.

Date	Plot no.	Rainfall (mm)	Runoff depth in collector_1 (cm)	Runoff depth in collector_2 (cm)	Fraction (X)	Runoff sample code	Remarks
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When bigger runoff plots are used, the size of the collector can't accommodate the runoff; therefore, multi-slot-divisor runoff collectors should be used (Figure 5). There should be two or more runoff collectors. One-ninth of the runoff from the first runoff collector and one-fifth from the second collector are directed to the collector tanks using slot divisors. The central slot divisors are connected to collection tanks through tubes. The runoff collected through the single-slot divisor in each tank should be multiplied by the number of slot divisors and, finally, the total runoff from all collectors should be summed up to calculate the total runoff depth from a plot.

Figure 5. Multi-slot divisor runoff plot.

Source: Adimassu and Haile, 2011.

5.3.6 Stream discharge (gauged and ungauged watersheds)

Runoff can also be measured at the outlet of the watershed using stream/river gauging, which is based on measurement of cross-sectional area and stream flow velocity (area-velocity methods) and stream water levels using weirs or flumes and automatic water-level recording instruments.

- 1. Area-velocity method:** Measure stream cross-section (width and depth of the stream). However, in most cases, the stream cross-section is not smooth; rather, it is rugged, so use surveying equipment to measure stream cross-sectional area more precisely. Flow velocity could be measured using a

floaters or current meter. Integrating the stream cross-section and flow velocity will help to establish stream stage and discharge relationships for predicting stream discharge (Table 18).

- 2. Flumes or weirs:** For small stream construction, install a rectangular weir, V-notch weir, or flume to measure depth and estimate discharge using the respective height discharge equations.
- 3. Automatic water-level recorders:** To measure discharges of small to medium rivers/streams, a diver and baro diver could be used. Readings could be taken at 5-min intervals although this could be modified depending on the climate, time of concentration of the watershed, and other requirements. The baro diver also records atmospheric pressure, which will be subtracted from the measurements of the diver measuring total (atmospheric and water) pressure at the bottom of the stream/riverbed. The data recorded by the diver and data logger can be downloaded to the computer using a cable or infrared readout to calculate the discharge rate.
- 4. Radar method:** For large rivers, radar could be used.

Table 18. Data for measuring stream discharge at gauged watersheds.

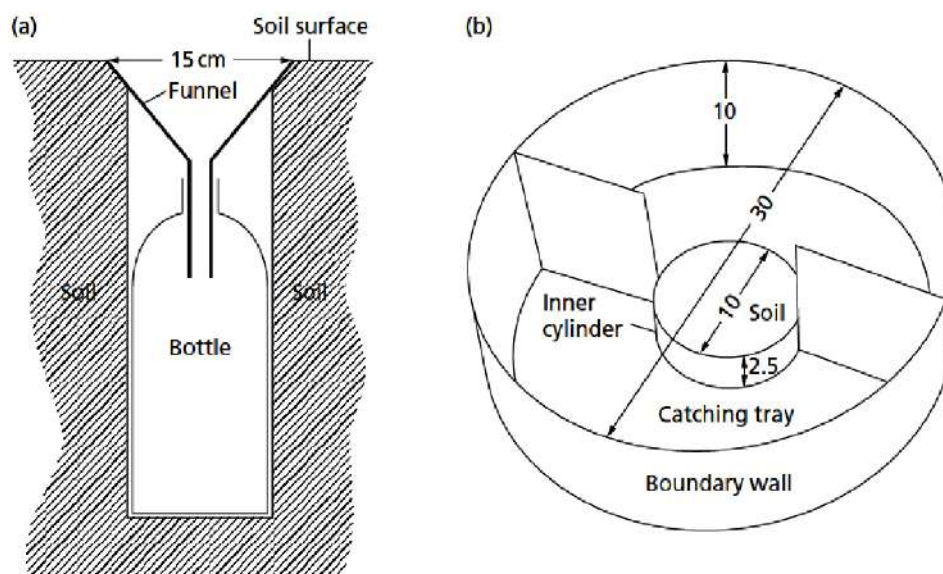
Date	Time	Stream width (m)	Average stream depth (m)	Stream flow velocity (m/s)
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5.3.7 Splash erosion

Raindrops that hit bare soil with full force can break apart the soil structure in a process called rain splash erosion. As raindrops strike the ground, they break down soil particles from the larger soil mass and scatter them in all directions. Loosened particles are then easily carried away by flowing water. This process is the primary source of soil particles available for erosion and transport during rainfall events.

Measuring splash erosion:

- Splash erosion can be measured in different soil types of varying erodibility. So, it is necessary to identify the major soil types in the watershed and locate where splash erosion measurement is to be conducted.
- Characterize the soil type, vegetation (type and coverage), slope, and related topographic features, rainfall characteristics, etc.
- Use a **splash board, cup, or small funnels or bottles** to measure rain splash erosion in the field (Figure 6).
- These devices are inserted in surface soil, but care should be taken to control the contribution of overland flow to the splash cup and funnel when measuring splash erosion.
- Splash erosion can also be measured by collecting detached and splashed soil materials or using rainfall simulators.
- The amount of soil or sediment material collected due to splash erosion depends on the diameter of the measuring device. Therefore, the measured data need to be corrected considering the diameter of the splash cup, board, or funnel (Morgan, 2005).

Figure 6. Examples of devices for measuring splash erosion: (a) splash funnel and (b) field splash cup.

Source: Morgan (1981, 2005).

- Use the relation $MSR = MS e^{0.054D}$ to estimate splash loss. MSR is mass of real splashed material per unit area (g/cm^2), MS is measured splash per unit area (g/cm^2), and D is the diameter of the cup and funnel (Morgan, 2005) (Table 19).

Table 19. Data format for measured splash erosion assessment.

S. no.	Date	Time	Rainfall depth (mm)	Rainfall intensity (mm/h)	MS (g/cm^2)	Diameter of cup (cm)	Rate of splash erosion (t/ha)	Remarks
1								
2								

5.3.8 Inter-rill erosion

Inter-rill erosion results from uniform removal of topsoil due to overland flow. This form of erosion is difficult to recognize in the field, but some indicators are plant root exposure, rock exposure, and change in soil color. Inter-rill erosion is measured using micro-plots ($3 m^2$) that represent the different soil types (with varying erodibility) and areas receiving different rainfall amounts and inclination/rainfall direction (Table 20). However, this measurement involving runoff plots includes the measurement of splash, sheet, and rill erosion in combination. To measure how much soil has been lost to inter-rill erosion:

- Use erosion pins installed in the soil, protruding above the soil and marked at the surface so that soil depth reduced due to inter-rill erosion can be determined.
- Erosion pins can be installed considering spatial variation in micro-topography, soil properties, and land cover/land use.
- Several replications (at least three pins) are needed for each unit of land and slope (MU).
- Soil depth reduction is recorded and the affected area is measured to estimate the total volume of soil lost.
- A 1-mm soil depth reduction corresponds to approximately 10 to 15 t/ha/year depending on soil bulk density.

Table 20. Data collection format for assessing field inter-rill erosion.

Date	Measuring/ mapping unit (MU)	Pin number	Pin depth (mm)	Area proportion of MU	Soil loss rate (t/ha/year)
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5.3.9 Soil erosion measurement using runoff plot

Runoff plots could be used to measure erosion due to combinations of erosion factors/processes. Use the following procedures:

- ▶ Use plots of 4 m × 15 m for <20% slope and 4 m × 10 m for >20% slope (representing most erodible soils; major land covers and slope classes).
- ▶ Select desired plot locations for the different soil types, vegetation cover, and topographic features.
- ▶ Care should be taken not to compact, disturb, and translocate the soil during plot installation, in particular around the plot border or boundary.
- ▶ Required instrumentation:
 - » Bounding material metal plate or wooden or compacted soil bund or concrete wall.
 - » Gutter or inlet tube to convey runoff water and sediment from the plot to storage.
 - » Collecting tank volume determined based on maximum runoff and sediment volume expected from the plot. The collector tanks can be two or more depending on the event runoff volume generated from runoff plots.
 - » Runoff and sediment sampling devices.
 - » Lab facilities to determine sediment concentration.
- ▶ Runoff sampling procedures for collecting runoff from the plot and sediment sampling procedure include the following:
 - » Mix the runoff with the sediment to ensure uniform distribution of the sediment and sample representativeness.
 - » Prepare runoff and sediment sampling devices.
 - » Collect 1-L-depth integrated runoff and sediment sample.
 - » Filter the sample to determine sediment concentration.
- ▶ Frequency of sampling depends on the purpose of measurement: storm-/event-based sampling or daily sampling. Daily sampling is usually enough for soil erosion measurement for watershed studies.
- ▶ Monitor changes in plot characteristics such as vegetation cover percentage, management such as cropping and crop growth stage, percentage ground cover, etc.
- ▶ Soil loss due to sheet and rill erosion from the plot is calculated based on the runoff volume, sediment concentration, plot area, and sampling duration.
- ▶ Uncertainties are related to soil erosion measurements. Therefore, measurement accuracy is important, considering the different sources of measuring errors.

Sediment sampling

The clear portion of the runoff collected in collector 1 (tank 1) should be siphoned out carefully. The remaining sediment must be stirred thoroughly and weighed together with sediment deposited on a ditch and a known amount of sediment sample taken for gravimetric moisture content determination (Tables 21 and 22). The samples must be oven-dried at 105 oC for 24 hours. The volume of runoff in collector 2 (tank 2) must also be measured, stirred thoroughly, and a 1-L sample taken for sediment concentration analysis. Sediment concentration multiplied by the total volume of runoff is used to estimate the total soil losses from each rainstorm.

Total soil loss (ST) is calculated as

$$S_T = S_D + S_A + S_B$$

where S_T = weight of total soil loss calculated

S_D = weight of dry soil loss collected from runoff and sediment collecting ditch

S_A = weight of dry soil loss recorded in tank 1

S_B = weight of dry soil loss calculated in tank 2

$$S_D = S_{DW} \times S_{DO}$$

where S_{DW} = weight of wet soil loss recorded on runoff and sediment collecting ditch

S_{DO} = oven-dried soil percentage of soil sample from runoff collecting ditch

$$S_A = S_{AW} \times S_{AO}$$

where S_{AW} = weight of wet soil loss recorded in Tank 1

S_{AO} = oven-dried soil percentage of soil sample in Tank 1

$$S_B = R_T \times R_C$$

where R_C = runoff sediment concentration in Tank 1

Instruments: Bounded experimental plot, slot divisor, runoff collecting tanks, sediment collecting ditch, measuring stick, graduated cylinder, sediment sampling bottles, filter paper, beakers, balance, oven, markers.

Table 21. Runoff and sediment sampling from runoff collectors connected to runoff plot.

Date	Time	Plot no.	Rainfall (mm)	Runoff depth (cm), Collector_1	Runoff depth (cm), Collector_2	Fraction	Total runoff (L)	Sample code	Sed. conc. (g/L)
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Table 22. Sediment sampling in runoff plots.

Date	Plot no.	Sediment concentration Tank 1 (g/L)	Dry soil loss in Tank 1 (kg)	Sediment concentration Tank 2 (g/L)	Dry soil loss in Tank 2 (kg)	Dry soil loss on ditch (kg)	Total soil loss (t/ha)	Remarks
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5.3.10 Rill erosion

In this guideline, we adopted Herweg's Assessment of Current Erosion Damage (ACED) to determine rill erosion dimensions and rill erosion hotspots. Data on rill dimension (length, width, depth), rill density and distribution, and rill damage area can be assessed using ACED, which has two major purposes:

1. supplement existing erosion measurement data at different levels, such as test plots and river gauging stations, and
2. provide practitioners with a more cost-effective tool to assess soil erosion and draw conclusions about implementing SWC measures.

Being a rough method of soil erosion assessment, ACED cannot provide the same level of accuracy as test plots or stream/river gauging. The level of accuracy can be $\pm 15\%$ and accuracy further decreases up to $\pm 30\%$ for less experienced researchers/experts, when vegetation cover percentage increases or when the rills become more complex. ACED or hotspot damage mapping involves point-linear measurement, measuring soil loss at critical locations during severe rainstorms. It is a short-term monitoring method and results often indicate hotspot erosion areas.

Procedures:

- ▶ Conduct a quick transect survey over the watershed.
- ▶ Identify areas with active rill erosion damage, concentrated flow, and other vulnerable areas, and geo-reference these locations.
- ▶ Divide the rills into uniform segments based on the cross-sectional area and depth of the rill.
- ▶ For each segment, measure the depth, width, and length and calculate the volume of the rill for each segment.
- ▶ Sum up the volume of all the segments and calculate the total volume of the rill.
- ▶ Since soil loss is reported in units of mass per unit area and time, the volume should be multiplied by representative soil bulk density to convert from the volume unit to the mass unit.

5.3.11 Gully and bank gully survey

In gully damage areas, source area, head cut and gully-head retreat, gully network/pattern, and segments of gully dimensions (length, width, depth) are parameters to be measured (Table 23).

Procedures:

- ▶ Divide the main gully into uniform segments at every 10-m interval depending on the uniformity of the gully in its width and depth.
- ▶ For each gully and bank gully segment, measure the width, depth, and length and calculate the volume (m^3).
- ▶ Sum up all the calculated volumes to obtain the total volume (m^3).
- ▶ Multiply the total volume by a representative soil bulk density to calculate soil loss in the mass unit.
- ▶ Since gullies may not be created due to annual rainfall, proper documentation of the age of the gully is important through discussions/interviews with the local community. This will help to establish long-term soil loss rate and factors triggering gullies in the watershed.

Table 23. Data format for recording gully or rill dimensions and volume and area of damage.

Segment number	MU	Width (m)	Depth (m)	Cross-sectional area (m ²)	Length (m)	Volume (m ³)	Representative bulk density (g/cm ³)	Soil loss (t/ha/year)
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1

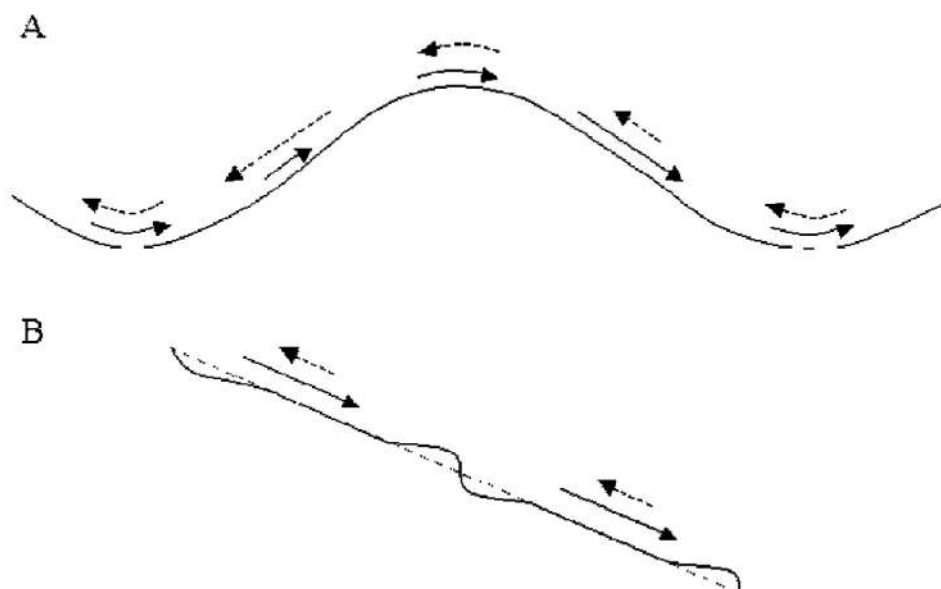
2

Note: For rill erosion, the measurement would be in cm.

5.3.12 Tillage erosion

The effects of water and wind erosion are often dramatic and can easily be identified in the field. However, the effects and severity of tillage erosion become apparent only after several decades of tillage operation and are less observable by farmers in the field. During tillage operation, soils are redistributed by tillage of various implements used in both mechanized and non-mechanized agriculture. The rates of soil translocation by tillage depend on tillage depth, type of tillage implement, slope, speed of tillage operation, direction and frequency of tillage, and soil properties. Tillage erosion can be of the same magnitude as water erosion. Tillage erosion operates like a conveyor belt, transferring soil and associated constituents from convexity to concavity (Figure 7). Tillage erosion is expressed in terms of tillage transport coefficient (Q_s), which is mostly constant and ranges on the order of 400–800 kg/m/year for mechanized farm implements and 70–260 kg/m/year for a non-mechanized implement.

Figure 7. Tillage erosion-slope relationship illustrating net movement of the soil downslope. Compare the upward and downward arrow length representing the magnitude of soil displaced.



Source: Vieira and Dabney, 2011.

Figure 8. Tillage erosion caused by traditional “arc” plow “maresha” and its significance in soil distribution.



Source: Nyssen et al., 2000.

Measuring tillage erosion

Nyssen et al. (2000) developed a method to calculate tillage erosion rate using a traditional “ard” plow in Ethiopia (Figure 8), and the same method is recommended, which includes the following:

- ▶ Identify areas where tillage erosion is to be measured.
- ▶ Number stones and hide them in the soil but measure the location of each of the stones from a given reference point.
- ▶ Allow the farmers to till their land normally (without knowing where these stones are).
- ▶ Find the new locations of the stones and measure the distances of these stones from the reference point.
- ▶ Calculate the distance that these rocks are translocated by tillage passes (Figure 9).
- ▶ This provides the rate of translocation of geological materials due to tillage.

5.3.13 Sediment yield

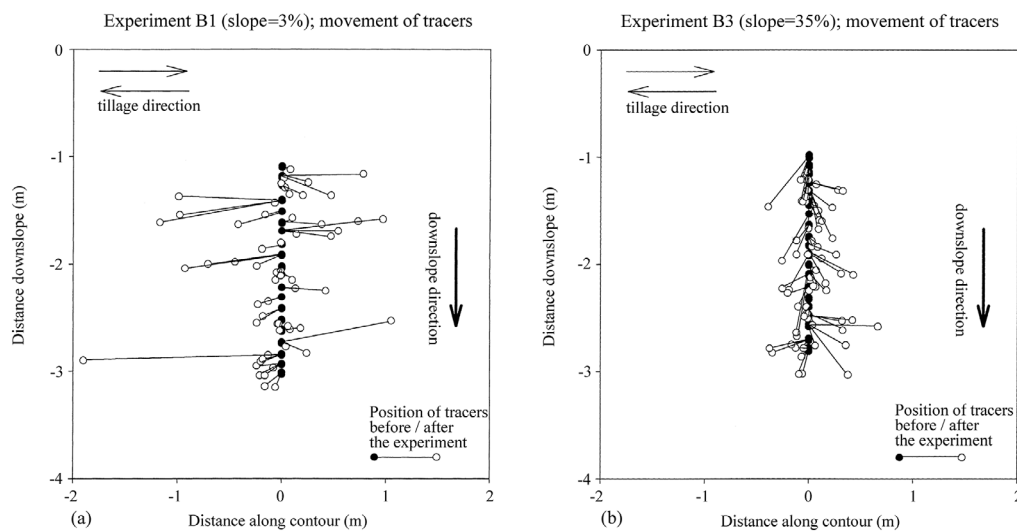
Sediment yield is the mass of sediment crossing a specific field boundary, watershed, catchment, or basin and is expressed in units of mass of sediment per unit area and time, which is also called specific sediment yield.

Sediment yield can be measured using

1. Fluvial monitoring
2. Reservoir survey
3. Application of a sediment yield model

Fluvial monitoring is a continuous process of measuring discharge and sediment concentration within the discharge. Data on sediment yield at the outlet of the river basin reveal the rate of erosion and soil loss of the upstream watershed.

Figure 9. Measuring tillage erosion rate using displacement of stones (rock fragment) to simulate soil displacement.



Sources: Nyssen et al. (2000, 2001).

i. Manual sampling strategies

Discharge and sediment concentration measurements using manual methods are very difficult for large streams or rivers characterized by flash flood. Discharge and sediment concentration measurement should consider the likely behavior of the river and the timing of significant concentration fluctuations. This is the case when rain is falling frequently and causing high flow fluctuation in the stream/river and fluctuation of sediment concentration.

The procedure for manual sampling involves measuring

- Water level (using a staff gauge)
- Stream velocity (using a floating or current meter)
- Discharge (using velocity area methods)
- Sediment concentration
 - » Water sampling (using manual sampling)
 - » Filtration (to extract the sediment)
 - » Analysis of grain size

ii. Automatic sampling equipment and monitoring

Upstream soil erosion and downstream sediment yield are linked/related to each other by sediment delivery ratio (SDR). The SDR is affected by the location of sediment source areas in the watershed, topographic characteristics, vegetation cover, and climate.

Procedure for automatic measurement:

- An automatic sampler is used to take runoff and sediment samples during a specific time interval.
- Sediment yield will be calculated from the discharge and sediment concentration (Table 24).
- The sampling scale for sediment yield measurement can be at any scale: micro-watershed, sub-catchment; the measurement accuracy error ranges from 5% to 10%.

Table 24. Data format for measuring sediment yield.

Date	Time	Sample code	Discharge (m ³ /s)	Sed. conc. (g/L)

5.3.14 Reservoir sediment yield survey

Artificial water reservoirs are constructed for irrigation development, drinking, aquaculture, hydropower, flood control, or combinations. However, the presence of reservoirs (natural or man-made) creates an excellent impoundment to trap sediment during transportation. Successive reservoir surveys provide information on the rate of soil erosion and sediment yield from the catchment of the reservoir. Standard methods exist for conducting reservoir surveys for the estimation of soil erosion rate from the catchment.

Overlay analysis of successive maps

For man-made reservoirs, it is necessary to acquire a topographic map of the reservoirs during the design phase and develop a Digital Elevation Model (DEM). Water depth could be measured using different methods, for example: (a) sonic depth recorders, (b) bathymetric surveys, and (c) grid-square method in dry reservoirs. Based on these data, a topographic map of the current topography of the reservoir could be constructed at the same scale as the original map. Digitize the same maps and develop a data layer between the two DEMs. The volume difference between the two maps indicates the sediment stored over the years, that is, between survey intervals, which can be calculated using GIS software. This method can be applied when the quantity of inflow and outflow from reservoirs is known in each measurement period.

The following options can be used:

a. Sonic depth recorders

Sonic depth recorders are used to measure water depth along predetermined ranges and locations. Sediment yield can be calculated from previous surveys of the original reservoir bottom or from the soundings of the sediment depth.

b. Area-elevation-capacity curve

The area and capacity of the reservoir are determined from the topographic map of the reservoir mapped during the design phase (Figure 10).

The area enclosed in each contour is measured with a planimeter.

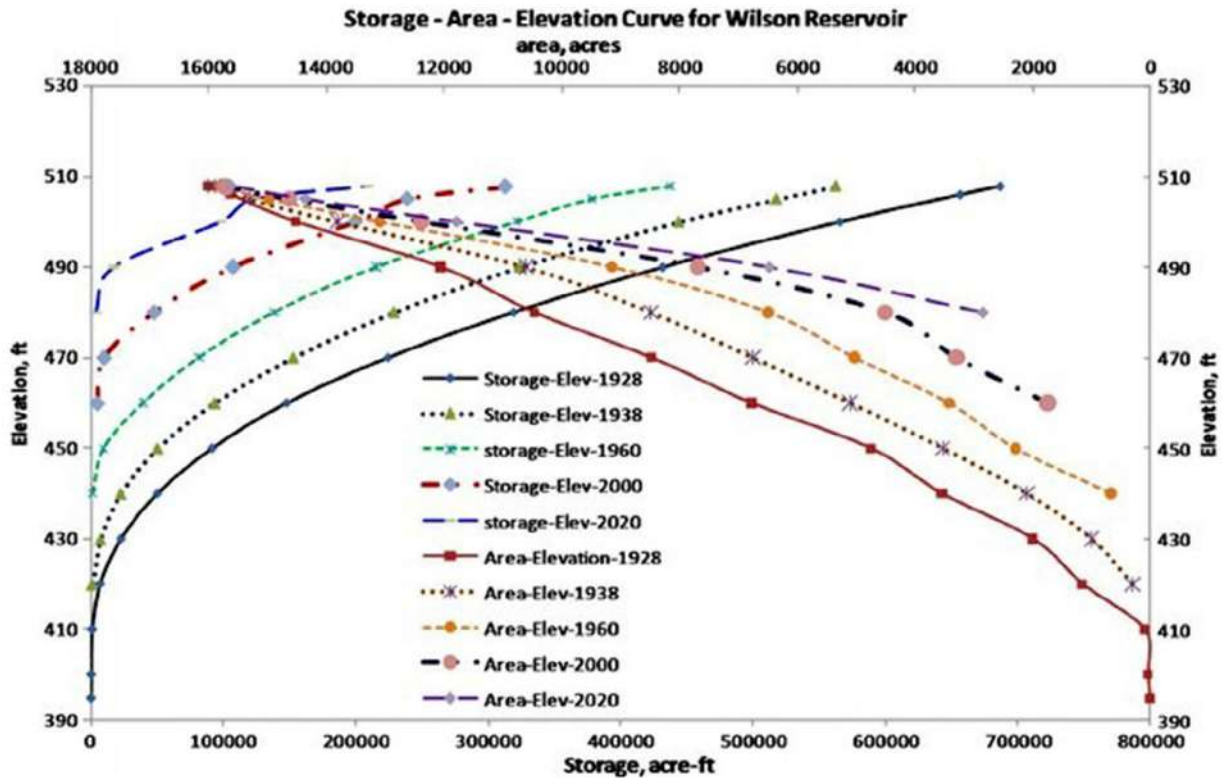
- » The decrease or increase in volume due to sediment yield stored and water yield decline or rise, respectively, between two successive contours is determined by using the average area method as follows:

$$V = \frac{(A1+A2)h}{2}$$

where A1 and A2 are areas of successive contours and h is height or elevation difference between successive contours.

- » The volume of sediment accumulated over the years will be determined by reading the volume corresponding to the current average sediment level in the reservoir.

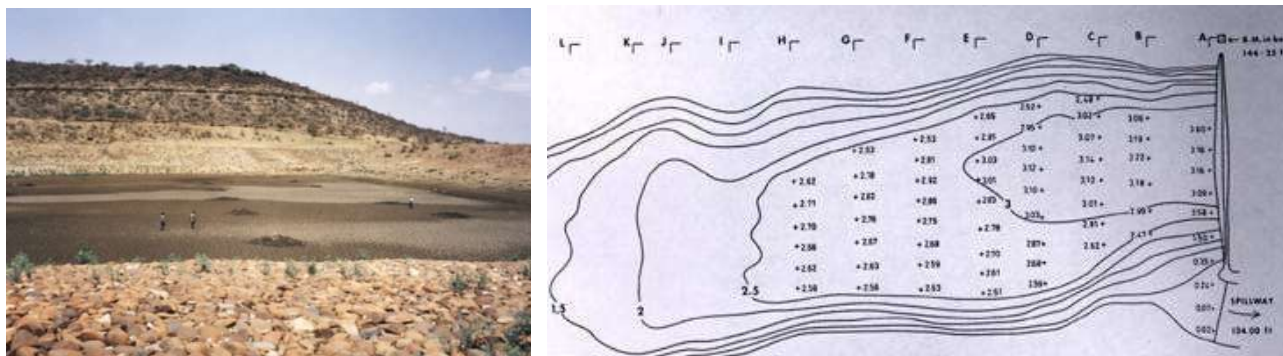
Figure 10. Reservoir elevation area and volume relation curve (Gebregiorgis and Hossain, 2012).



c. Grid-square method

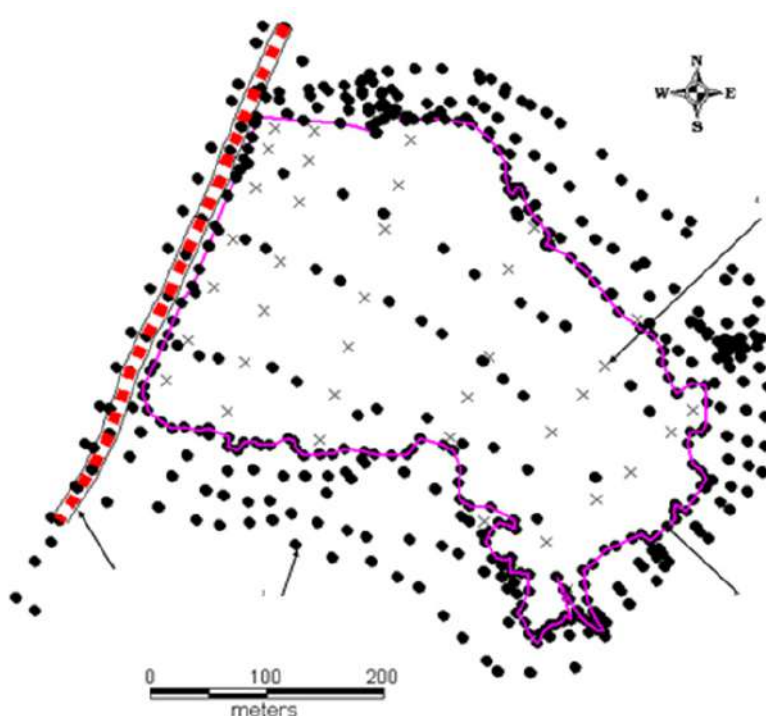
- » For reservoirs that become dry at the end of the dry season, the water stored in the reservoir is usually completely used for irrigation and livestock consumption. Therefore, it is possible to measure sediment thickness by using regularly spaced profile pits or by auguring.
- » The volume of sediment could be calculated with the following procedures:
 - Measure sediment thickness using meter sticks by opening profile pits per reservoir along transects (Figure 11).
 - The number of pits will be determined by the size and nature of the original ground level of the reservoir. If the original ground level is rugged at the time of construction, the thickness of the deposited sediment is also expected to vary.
 - Map deposited sediment boundary, location of pits (Figure 12), and topography of the current ground level (bottom of the reservoir) using surveying instruments (theodolite or others such as Total Station).
 - Delimit the boundary of the sediment deposition area (thickness = 0) by digitizing the surveyed points as a polygon in IDRISI software.
 - Produce a DEM in GIS with X and Y readings taken using surveying instruments, taking measured sediment thickness as the z value.
 - Sediment volume is then determined by running the EXTRACT module in IDRISI®, which extracts summary statistics from an existing image file (Haregeweyn et al., 2006; Tamene et al., 2011).

Figure 11. Measuring sediment thickness by opening profile pits along transects.



Source: Haregeweyn et al. (2006); Tamene et al. (2011).

Figure 12. Reservoir survey using grid-square method showing (1) dam, (2) theodolite reading points, (3) deposited sediment boundaries, and (4) location of sediment profile pits.



Source: Haregeweyn et al. (2006); Tamene and Vlek (2007).

5.3.15 Wind erosion

Wind erosion is a process in which powerful air currents lift soil particles from the ground and carry them to new locations. Transportation by wind can be in the form of suspension, saltation, or surface creep depending on particle size. Large particles greater than 500 μm are transported by surface creep, rolling in contact with the ground surface, whereas intermediate particles (50–100 μm) are transported by saltation or series jump, and small particles in suspension.

Wind erosion occurs when the soil is dry, is low in organic matter content, and has low vegetation cover and surface roughness. In Ethiopia, although water erosion processes such as sheet, rill, and gully are important in the central and north highlands, wind erosion is important in the peripheral lowlands of the country, mainly in the Rift Valley and rift escarpment. Erosion science for wind erosion and its measurement methods are limited in Ethiopia.

Measuring wind erosion:

- ▶ Sand drift can be measured using saltation traps.
- ▶ Measurements of wind and dust concentration profiles can be conducted using anemometers to indicate wind direction and dust samplers mounted on towers (Figure 13).
- ▶ Wind erosivity and wind erodibility can also be used for assessing wind erosion.
- ▶ Based on surface factors (surface roughness) and wind-related factors (wind speed and turbulence), the factors in wind erosion can be modeled.
- ▶ Wind erosivity describes the potential of wind to generate erosion or dust storms, while erodibility describes the potential of the surface to be eroded.
- ▶ Wind erosion indices can be developed with the data of wind speed, precipitation, and evaporation.

5.3.16 Soil moisture content

Soil moisture content plays a vital role in watershed productivity by directly impacting agricultural yield, livestock grazing capacity, and forest health. This moisture content influences hydrological, ecological, climatological, and atmospheric watershed processes.

To accurately measure soil moisture content across a watershed (Table 25):

1. Divide the watershed based on land use, slope, geology, and soil characteristics.
2. Collect soil samples through either
 - » Gravimetric sampling: taking 100–200-g soil samples every 20-cm depth and weighing before and after oven-drying at 105 °C for 24 hours without a cover to remove all water in the soil (Taye et al., 2024).
 - » Electronic monitoring: using TDR probes and data loggers for real-time moisture tracking at various depths.

Figure 13. Wind erosion and dust sampling methods.



Source: Meng et al., 2024.

These measurements enable modeling of soil moisture distribution patterns and their changes over time, thus providing crucial data for watershed management and agricultural planning.

Percentage soil moisture can be calculated using the following equation (for detailed information and other methods, see Hudson, 1993):

$$\text{Soil moisture (\%)} = (\text{wet soil} - \text{dry soil}) * 100 / \text{dry soil}$$

Instruments: Augur, airtight container or plastic bag of known weight (cane), balance, and oven.

Table 25. Data format for recording soil moisture using a gravimeter.

Sampling date	MU	Sample depth (cm)	Sample code	Wet weight (g)	Cane weight (g)	Dry weight (g)	Soil moisture content by weight (%)	Bulk density (g/cm ³)	Soil moisture content by volume (%)
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5.3.17 Infiltration

Infiltration is the most difficult parameter to determine accurately and needs considerable time for determination (Table 26). In determining infiltration capacity of soils, it is necessary to consider variation in soil infiltration rates as well as groundwater conditions along landscapes. Common techniques for measuring infiltration are (a) double ring infiltrometer, (b) mini disk infiltrometer, (c) (inverted) auger hole methods, and (d) micro-plot methods. Other methods include ponding, blocked recirculating infiltrometers, and the tailwater hydrograph. (For detailed information, refer to Walker, 1989)

Table 26. Data collecting forms for infiltration.

Date	Plot/location	Time (h:min:s)	Water level (mm)	Remarks
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5.4 Data standard for watershed technology evaluation and impact study

5.4.1 Evaluating quality and efficiency of SWC practices at watershed scale

Purpose: Quality assessments evaluate how well soil and water conservation (SWC) meet design standards and conservation goals. This monitoring provides data on their effectiveness in reducing soil loss, managing watershed sediment, and enhancing groundwater recharge. The results identify maintenance needs and validate the impact of conservation practices.

Sampling: To evaluate the quality and efficiency of SWC practices or technologies (mechanical and biological) at watershed scale, divide the watershed into homogeneous sub-catchments based on factors that affect the placement and design of SWC measures such as land use, soil drainage/texture, slope, landform, and geology.

To understand upstream and downstream connectivity or interaction of SWC practices with slope profile, homogeneous sub-catchments can be further divided into upper, middle, and lower landscape positions.

Sample at least three consecutive SWC measures at each position (bunds, trenches, drainage ditches, hedgerows, or grass strips, etc.). Measure design parameters (length, width, height, depth, gradient), layout parameters (spacing, vertical interval, density/count), sediment retention (depth and volume), and age and level of damage to structures, and take into account maintenance measures. Take spacing/vertical interval data in each sample structure to estimate linear SWC structure density.

Methods: To measure sediment retention of bunds, detailed sampling along the structure is required. Sample three consecutive structures per landscape position (Table 27). Mark 30 m in length on each structure and take measurements at three points at 10-m intervals, measuring selected parameters such as dimension, sediment accumulation, and level of damage. Measure five times during the rainy season: twice early immediately after crop planting, twice mid-season, and once at the end of the season.

Accumulated sediment represents a portion of soil loss from the terraced area that would otherwise flow downslope. To measure the amount of sediment accumulated/retained by the structures, two independent methods can be used:

1. For structures with runoff drainage ditches on the upper side (e.g., graded soil bunds and stone bunds).
2. For those without runoff ditches (e.g., fanya juu, grass strip, bench terrace).

For structures with ditches, sediment accumulation is measured along the ditch on the upper side of the bunds. In each sample point at 10-m intervals of 30 m in total length, first we assume/estimate the original dimension of the ditch (depth and width) and then measure the depth of the remaining top part of the ditch (i.e., depth above the accumulated sediment) and subtract it from the original depth of the ditch to estimate the amount of sediment retained or accumulated on ditches. This amount is a fraction of the soil loss washed from the terraced area and part of the total soil loss washed down and transported out along the gradient. The volume of sediment accumulated per unit length of structure can then be estimated by multiplying the depth of the sediment by the cross-sectional area of the ditch.

For those structures without a ditch, sediment retained/accumulated along the embankment of structures could be measured by either of the following two methods:

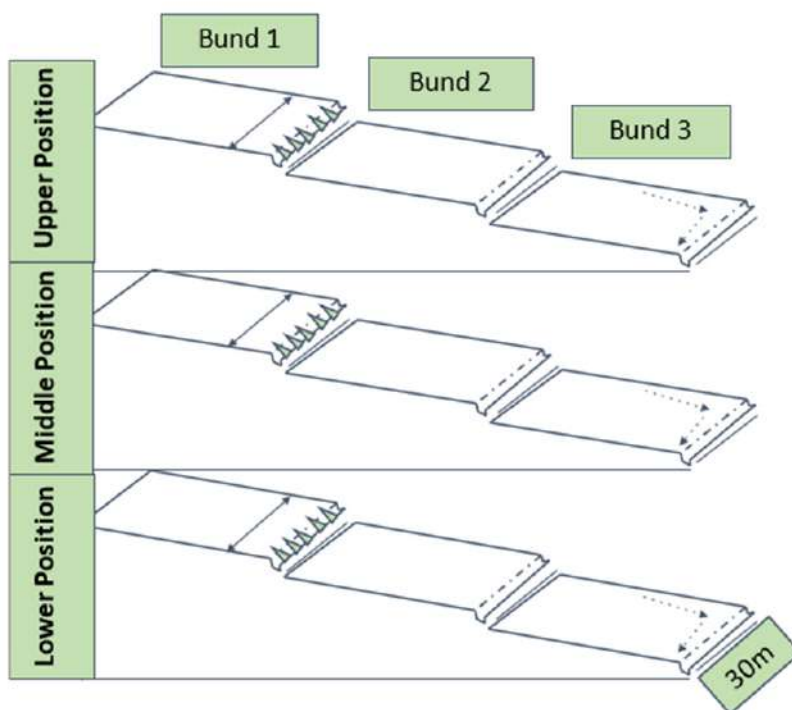
1. Before the start of rain, lay out a 1-m plastic sheet at three points (center of 10-m intervals) along a 30-m-long sediment accumulation area. Measure the volume of sediment using buckets or weight of sediment every 2 weeks until the end of the rainy season.
2. Before the start of rain, install enough graduated/marked wooden pegs or iron pins at 10-m intervals along the 30-m length (with a minimum of three pegs per interval). Measure the depth of accumulated sediment every 2 weeks until the end of the rainy season. The volume of sediment retained is estimated by multiplying the average depth by the sediment accumulation area (Tables 28 and 29).

In cases where baseline information is not available on sediment accumulation efficiency of structures that were constructed some years back, we can reconstruct the slope profile change (before construction of bunds and after some years) induced by SWC structures in each landscape position (upper, middle, and lower) (Figure 14). We can then estimate the sediment retained per year and change in sediment retained between two periods.

In addition to the above method, which could be used to roughly estimate the distribution and extent of SWC implementation, it is more appropriate to apply Google Earth to digitize and estimate the actual density of SWC structures. Thus, to estimate the total length of linear structures (terraces, waterways, cut-off drains, etc.) constructed in a watershed, high-resolution Google Earth images can be used to digitize each structure onscreen and work out the density of the structures (length per unit area) in a watershed. Similarly, linear erosion features such as gullies, rills, and riverbanks as well as landslide

damage areas and the associated extent/area coverage can be digitized using Google Earth images. Point features such as water points and other infrastructure can be either digitized using Google Earth or measured in the field using GPS.

Figure 14. Slope profile change induced by SWC structures can be reconstructed in upper, middle, and lower landscape positions.



Source: Authors' construction.



Table 27. Format for measuring and documenting the dimensions of soil and water conservation structures.

Watershed	Sub-catchment name	Slope position	Sample bund	Type of bund + veg.	Age (year)	Spacing b/n bunds (m)	Sample point (1 to 3 from top down)	Height (cm)	Height of embankment (cm)	Width (cm)	Length (m)	Gradient (%)
		Upper	1	Soil bund			1					
			1				2					
			1				3					
			2				1					
			2				2					
			2				3					
			3				1					
			3				2					
			3				3					

Table 28. Format for measuring sediment volume retained by SWC structures (Sed-RT) – Method 1 by measuring ditch dimensions.

Watershed	Sub-catchment name	Slope position	Sample bund	Type of bund + veg.	Sample point (1 to 3 from top down)	Orig. depth (cm)	Ditch width (cm)	Depth from top of ditch (cm)	Depth of sediment (original depth minus depth from top) (cm)	Spacing between bunds (m)
		Upper	1	Soil bund	1					
			1		2					
			1		3					
			2		1					
			2		2					
			2		3					
			3		1					
			3		2					
			3		3					

Table 29. Format for measuring volume of sediment retained by SWC structures (Sed-RT) – Method 2 using plastic sheet or Method 3 using wooden pegs/iron pins.

Watershed	Sub-catchment name	Slope position	Sample bund	Type of bund + veg.	Sample point (1 to 3 from top down)	Method 2		Method 3		
						Vol. of sediment in bucket (L)	Weight of sediment (kg)	Average depth/peg height (cm)	Sed. accumulation area (m ²)	Vol. of sediment (m ³)
		Upper	1	Soil bund	1					
			1		2					
			1		3					
			2		1					
			2		2					
			2		3					
			3		1					
			3		2					
			3		3					

5.4.2 Evaluation of SWC technologies using experimental plot

This guideline uses SCRP experimental plot setup and soil loss and runoff data collection methods. The effects of soil conservation measures or land management practices on soil loss, runoff, soil fertility, crop yield, and biomass production can be tested against control plots representing local practices without SWC measures vs experimental plots (EP) of 180 m² (6 m × 30 m) each. This plot size has been found to be more appropriate in Ethiopian highlands, where land size is more fragmented. Under very distinct conditions such as in the dryland and low-rainfall areas where the slope is relatively flat, allowing for wider spacing between structures/management practices, and when comparing several management practices (for example, conservation agriculture practices), the plot size can be flexible, with at least three conservation structures/bunds included. In flat areas, a plot size of 4-m width by 45-m length could be adequate.

Methods: Measurements can be taken of daily runoff and soil loss after rain for plots with different SWC measures or practices (Table 30). Each plot is bounded by corrugated iron borders inserted 10 cm in the ground and 20 cm above the ground to control splash loss and lateral flow, thus enabling runoff to be collected in the tank through an inlet tube. Plots are established near the rainfall recorder to link the results of rainfall and runoff periods. Runoff and sediments are collected in two 200–250-L tanks (Figure 15). In most of the cases, a multi-slot divisor (with 9 slots) was used where 1/9 of the runoff entered to the first tank. When the first tank filled up, runoff flow to the second tank. The volume of runoff water is measured directly from the depth of runoff in both tanks and the total runoff is nine times the runoff volume collected from the two tanks. Representative runoff and soil samples are taken for laboratory analysis. To measure soil losses, collect 500 g of soil and 1 L of suspended sediment samples from tank A and the overflow tank B, if necessary. The collection system is cleaned after each sampling. The 1-L sample, taken for sediment assessment, is filtered and dried. The 500-g sample taken from each plot is also dried in the oven and weighed.

Figure 15. Measuring daily runoff and soil loss from plots with the use of sedimentations tanks.



Source: Adimassu et al., 2014.

Table 30. Format for recording and documenting runoff volume and soil loss data from experimental plots.

Date	Plot no.	Time	WD A1	WD A2	WD A3	WG W1	WG W2	WG W3	Bucket W	Wet sample W	WD B1	WD B2	WD B3	OW A	Empty W	OW B
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5.4.3 Evaluating ex situ water harvesting (design, type, storage, functionality)

The design and performance evaluations of water harvesting structures require (a) proper evaluation of the amount of available water resources (surface water and groundwater), (b) selection of appropriate water harvesting/development technology that suits the local context, (c) selection of type of water resources, and (d) evaluation of the performance/functionality of the structures (Table 31).

Table 31. Data needed and parameters to be assessed in order to evaluate water harvesting systems.

No.	Parameter	Description
1	Location and date of construction	Name of place
		Latitude, longitude, elevation
		Date of construction
2	Rainfall, catchment characteristics, and runoff	Rainfall amount
		Rainfall intensity
		Catchment characteristics
		Volume of excess runoff
3	Available water sources	Surface runoff (runoff coefficient)
		Groundwater
4	Suitability of the site for the proposed water harvesting structure	Geology/geotechnical engineering
		Topography
		Storage capacity
		Any construction risks
5	Functionality	Siting problems
		Construction issues
		Inadequate maintenance
		Social issues

Continues ➤

No.	Parameter	Description
6	Challenges/problems	Siltation/sedimentation
		Erosion (surface/subsurface)
		Seepage
		Structural damage
		Other defects

Note: This information should be integrated with geology, hydrogeology, geomorphology, and land use/cover attribute maps and data.

5.4.4 Groundwater and spring water (water level, yield, and quality)

Effects of watershed management include groundwater recharge, enhanced spring discharge, and changes in water quality. To quantify impact, it is necessary to evaluate both groundwater and spring water before and after interventions. Regular monitoring requires groundwater measurements (static water level and quality of groundwater) and spring water measurements (discharge rate and water quality). The parameters to be evaluated appear in Table 32.

Table 32. Data and parameters needed to be assessed to evaluate and monitor groundwater and spring water.

No.	Parameter	Description
1	Location	Name of place
		Latitude, longitude, elevation
2	Drilling data	Date of drilling/development
		Depth of borehole
		Borehole diameter
3	Aquifer type and yield	Unconfined
		Semi-confined
		Confined
		Groundwater flow direction
		Well yield (L/s)
4	Groundwater	Method of measurement
		Measured date
		Static water level (m)
5	Spring water	Quality: physical, chemical, biological
		Discharge (L/s)
6	Frequency of static groundwater level measurement	At least twice a year: at the highest level and lowest level
		Before and after interventions

Continues ➤

No.	Parameter	Description
7	Water (groundwater and spring water) quality parameters to be determined	Physical
		Chemical
		Biological
8	Frequency of water quality measurement	During well and spring development and after development (at least twice a year, during high and low levels)
9	Functionality of the well	Siting problems
		Construction issues
		Inadequate maintenance
		Social issues

5.4.5 Crop and fodder productivity

Information on the effects of watershed interventions on crop productivity needs to be collected to determine the impact of SWC practices. This information can be sampled at different parts of the watershed representing heterogeneity in slope, soil, altitude, and crop type.

- **Sampling and crop yield data collection:** Seasonal (*Meher* and *Belg*) harvest samples shall be collected for all major crops in the watershed. Sample areas must represent various cultivation conditions (different slopes, soil types, and altitudes) and be chosen from fields with average crop stand conditions. The sampling will be conducted from at least 150 fixed plots and 150 non-fixed plots.

Base maps and farmer parcel maps can be used to guide the sampling process. Each sample plot will be assigned a unique plot number and marked on the parcel base map.

To evaluate the effect of SWC measures on crop productivity, harvest samples will be collected from three positions relative to field bunds: (1) position a – above the bund, (2) position b – between two successive bunds/terraces, and (3) position c – below the bund.

At each position, all plants, including weeds, will be collected in either a 3 m × 3 m or 2 m × 2 m area (Taye et al., 2024).

The plants will be harvested and grains with covers will be separated by hand and sun-dried for about 20 days. Straw will be weighed and recorded separately for each sample area.

Dried grains will be separated from covers by hand and final grain weights will be recorded on the harvest sample form per sample area. The grains and straw will be given back to the field owners.

Additional data collection includes documenting agronomic and crop cultivation information such as planting time, previous crops, and tillage frequency. Crop yield data (grain and biomass) will be collected, organized, and presented according to year, season, crop type, and intra-terrace areas/slope positions.

- **Monitoring fodder biomass production:** Samples of biomass harvest of fodder on bunds from 5-m length of bunds and 2 m × 2 m sample area from area closures, treated gullies, and grassland will be collected two times yearly (in two seasons, June and October) (Tables 33 and 34). Fresh weight will be directly measured and recorded.

Plow B: frequency of plowing before sowing

Plow A: frequency of plowing after sowing

Fertilizer: type of fertilizer applied

Sowing: sowing date and seeding rate

Croplast: crop of previous season

Croptwosa: crop two seasons ago

Depth (%)	Soil depth (cm)	Graincov (g)	Grain (g)
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Following	Crop (last)	Crop two seasons ago
-----------	-------------	----------------------

vegetation and restoring degraded
restoration strategy implemented
strategy.

regeneration, species composition, data are collected at different locations: on areas to evaluate growth rate, and diversity, and distribution of tree species.

s, can be used to collect vegetation
along each line transect, landscape
the number of transects and transect

intervals depends on the objective and heterogeneity of vegetation patterns at the site. Transects are laid out parallel to the topography. At each landscape position, quadrats measuring 20 m × 20 m are used and placed at certain intervals (to avoid edge effects of other land uses).

Growth parameters

In each of the quadrats (20 m × 20 m), individual plant variables such as count/survival rate per season, diameter at breast height (DBH) for smaller and multi-stemmed shrubs, diameter at stump height or at a height of 30 cm (d 30) from the ground, crown diameter, and total height of all woody species can be measured using diameter tape and clinometers, respectively.

Richness and diversity

The identity and abundance of seedlings and saplings can be identified in the field using the published volumes of the Flora of Ethiopia as well as Flora of Ethiopia and Eritrea and counted from subplots with 5 m × 5 m (at least two subplots) in the main quadrats. For those difficult to identify in the field, specimens are collected, processed, and identified by comparing them with already identified specimens deposited in the National Herbarium of Ethiopia. The types of plant species found at the study sites are also described in terms of richness, plant family, and habitat. The cover of herbaceous species is estimated in five subplots of 1 m × 1 m in the four corners and center of the main quadrats of 20 m × 20 m. To determine light availability, canopy openness can be measured using a densitometer positioned in the center of each quadrat.

The average total density of plant species per hectare is derived from the total number of individuals recorded in the sample plots. Sorensen's similarity index (Sorensen, 1948) is used to determine species similarities. The Shannon-Wiener diversity index (H' ; Shannon, 1948) is commonly used to determine native plant species diversity and to conduct comparisons in species diversity among the studied sites. To determine similarity in abundance of different species within a community, we can use the Shannon-Wiener index of evenness (J'). The regeneration capacity of restored sites for the trees and shrubs can be detected from soil seed bank samples and the regeneration inventory of species in the field.

Aboveground biomass and carbon estimation

To estimate aboveground biomass, inventory data are used to identify dominant woody species. Then, the methods of Hoff et al. (2002) and Snowdon et al. (2002) for aboveground woody biomass measurement are used. We advise grouping the dominant woody species into diameter classes to minimize errors that can arise from variable sizes of individual tree/shrub species. From the different land use sites, selected individual species representing dominant species can be harvested and weighed. Fresh mass of aboveground vegetation is then adjusted to dry mass using the measured moisture content, determined by oven-drying samples at 65 °C until constant mass is attained. Carbon stocks in aboveground biomass are estimated by multiplying the oven-dried biomass by a factor of 0.5 (Snowdon et al., 2002).

5.4.7 Unintended effects of watershed management

Watershed management changes runoff patterns through enhancing recharge and constructing different water harvesting structures. These dynamics could cause unintended effects, such as gully development, mass movements/landslides, flooding, siltation and seepage, and even infrastructural damage in the catchment/watershed. Watershed management could also result in unintended benefits that were not considered during the design and implementation phases. It is necessary to look at the interactions/dynamics in the whole catchment and possible unintended effects (negative and positive) of watershed management and water harvesting (Table 35).

Table 35. Parameters to be assessed in order to evaluate possible unintended (negative and positive) effects from integrated watershed management.

No.	Parameter	Description
1	Location	Place name
		Latitude, longitude, elevation
2	Surface erosion (gully)	Length
		Width
		Depth
3	Subsurface erosion (piping)	Length
		Width
		Depth
4	Mass movement (landslides)	Type
		Width
		Length
		Depth
5	Siltation/sedimentation	Farmlands in the valley bottom
		Water harvesting structures
		Settlement areas
		Roads and other infrastructure
6	Flooding	Farmlands
		Water harvesting structures
		Settlement areas
		Roads and other infrastructure
7	Any unintended benefits	

5.4.8 Costs of SWC structures/interventions

Watershed-based interventions should be sustainable in terms of social, economic, and environmental aspects. To enhance sustainable adoption and spread, introduced technologies and practices should be affordable to watershed users. Thus, it is important to know the costs of implementing an SWC technology or any management practice to estimate its benefits/profits. WOCAT questionnaires can be used to record the costs of technologies (WOCAT, 2016). If possible, break down the costs of establishment/initial investment and maintenance/recurrent costs according to Table 36. If you are unable to break down the costs, estimate the total costs of establishing/maintaining the technology (Table 37). Costs and inputs specified should refer to the actual technology area/technology unit applied in a watershed.

Table 36. Costs of establishment of technologies (initial investment cost) for integrated watershed management.

Input	Specify input ³	Unit ⁴	Quantity	Cost per unit	Total costs per input	% of costs borne by land users
Labor						
Equipment						
Plant material						
Fertilizer						
Construction material						
Others						
Total costs of establishment of technologies						

Table 37. Costs of inputs and recurrent activities needed for maintenance/operation (per year).

Input	Specify input ³	Unit ⁴	Quantity	Cost per unit	Total costs per input	% of costs borne by land users
Labor						
Equipment						
Plant material						
Fertilizer						
Others						
Total costs for maintenance of technologies						

3 Specify inputs. Labor includes total person-days, be they paid or unpaid (e.g., contributed by family members) or free labor contributions by the community. Under "Cost per unit," indicate daily wage for hired labor. If relevant, differentiate between skilled and unskilled labor. Equipment includes tools, machine hours, animal traction, etc. The cost calculation for machine hours and animal traction should be based on hiring costs – even if the machinery/animals are owned by the land user. Plant material includes seeds, seedlings, cuttings, etc. Fertilizers and biocides: compost/manure, inorganic fertilizer, herbicides, pesticides, etc. Construction material includes timber, stones, earth, cement, pipes, tanks, etc.

4 Units: person-days, kg, L, pieces, etc.

5.4.9 Community participation in monitoring and impact assessment of integrated watershed management

An appraisal of different watershed technologies and interventions in terms of adoption and effectiveness vis-à-vis local interests must be considered in the standardization. It must consider not only the technological means involved but also the approaches that are intended to facilitate successful implementation and adaptation of technologies. In evaluating problems, stakeholders such as farmers must be invited to take part in a broad participatory process to understand and analyze problems and express and evaluate their needs, interests, and aims. To enhance action-oriented research and development, continuous assessment and analysis of local community perceptions, farmers' decision-making processes, and the nature of farmers' responses to the respective indicators are important. On the other hand, land

management practitioners must first listen and learn; second, they must choose and match what they have to offer – technologies or professional advice – with people’s socioeconomic needs. Finally, they must monitor and evaluate what happens.

Farmers’ participatory watershed assessment involves the use of farmers’ knowledge base and farmer-expert joint learning for technology evaluation and improvement, adoption, and extension services. The procedure is designed to integrate the knowledge, attitudes, and preferences of farmers.

The farmer participatory approach is useful to

- ▶ Provide an opportunity for integrating technical solutions with farmers’ preferences and local knowledge.
- ▶ Facilitate farmer-participatory planning and implementation.
- ▶ Facilitate the acceptance, integration into the farming system, transfer, and application of technical findings and recommendations.
- ▶ Persuade farmers to adapt technologies without external enforcement.
- ▶ Share lessons and innovations from innovative farmers through popularization.
- ▶ Enable empowerment of the community in environmental knowledge and enhance bottom-up communication.
- ▶ Minimize the burden of development agents as farmers play the role of extension agents.
- ▶ Help to minimize farmers’ sense of dependency in conserving their land.

The approach involves the following methodological procedures:

- ▶ **Self-confidence-building measures:** This is related to awareness and attitude change activities to motivate farmers to engage in genuine participation and build trust. Self-confidence-building measures are carried out in the form of questions and answers.
- ▶ **Facilitation role of researchers and extension agents:** Extension agents and researchers play a role in facilitating motivation and exchange of knowledge and principles. They avoid a top-down imposition of technical knowledge. Moreover, kebele administration bodies help to facilitate and organize farmers’ interests.
- ▶ **Team/group formation and participation of land users:** Participation is such that all farmers are involved in the key collective decision-making processes. In addition, if it is difficult and unmanageable to involve all farmers, selected farmers’ teams can be organized to participate actively and engage in decision-making throughout the processes. In both cases, farmers are active leaders of the participation process. Watershed user cooperatives should be established to facilitate the tasks. Empowering cooperatives will lay the foundation for the exit strategy of any project or program.
- ▶ **Practically oriented knowledge sharing and upgrading:** Farmers are involved in exploring their practical experience and knowledge about watershed-level processes and interactions, drivers, and impacts through periodical meetings, field visits, and subsequent discussions; monitoring and measuring necessary processes; and evaluating interventions. All land users should be involved in one-to-one and group visits, discussions, and implementation activities on their own farms and adjacent farmlands.
- ▶ **Integrating assessment of field and toposequence units:** In smallholder farming systems, decision-making begins on the individual farm with the individual farmer as the starting point for planning. Farmers are asked to assess the watershed starting from their own fields; upstream and downstream of their farms (i.e., adjacent farms); and then for sub-catchments and watersheds. This means that understanding land and water degradation processes begins with an assessment at the individual farm level and ends with the catchments following the flow of water. Assessments at field scale begin

to evaluate the relative susceptibility of individual farmers and to identify the sources and causes of the degradation processes. The objectives of integrating field and toposquence assessment are to gain an understanding of the interrelations of the process-cause-impact sequence of processes and determine a collective understanding of constraints and solutions.

- ▶ **Recording explicit local knowledge and lessons learned:** This process helps convert implicit local knowledge to explicit scientific or practical knowledge. It helps develop articulated knowledge from local farmers' participatory process. It is the role of the facilitator researcher/expert to document the process and knowledge exchange and record changes in attitude, behavior, knowledge, and skill (implicit to explicit knowledge) at every stage.

6. Advances in Standardized Data Collection for Integrated Watershed Management

6.1 Introduction

Integrated watershed management is essential in sustainably managing water, soil, and other natural resources. Although the effectiveness of IWM depends heavily on data quality and consistency, developing countries have historically struggled with limited resources, outdated methods, and inconsistent measurement methods. The emergence of cost-effective technologies and standardized practices, however, is transforming how data are collected, shared, and used.

This section explores how technological advances (e.g., low-cost devices – free satellite data, mobile tools, and open-access platforms – along with improved data standardization practices) can be mobilized in developing countries to create a strong foundation for sustainable watershed management.

6.2 Affordable IoT solutions for soil and water monitoring

The landscape of hydrological data collection has evolved significantly. Where expensive equipment once created barriers for many regions, affordable Internet of Things devices have now democratized access to real-time environmental monitoring. These devices, including low-cost weather stations, portable water quality sensors, and soil moisture probes, provide data on key parameters such as rainfall, streamflow, groundwater level, and soil moisture content.

OpenWeatherMap exemplifies this shift by providing affordable weather data access while community initiatives such as the **EnviroDIY Mayfly Data Logger** enable local monitoring at significantly reduced costs (Musah et al., 2022). IoT networks facilitate remote data access, thus decreasing the financial burden of frequent field visits to remote or rural areas. Through consistent calibration and standard reporting protocol, these tools ensure reliable and comparable data across various regions (Hossain et al., 2021).

6.3 Using satellite imagery for watershed monitoring

Satellite imagery has become a critical watershed management tool. Free and open-access platforms such as **Sentinel-2** (European Space Agency) and **Landsat** (NASA) provide consistent and standardized information on land use, vegetation cover, and water bodies. These tools are particularly beneficial for developing countries with limited budgets for high-resolution spatial data. Platforms such as **Google Earth Engine** enable stakeholders to analyze satellite images without requiring advanced computing infrastructure. Indices such as NDVI (Normalized Difference Vegetation Index) can track vegetation

health, while DEMs from sources such as **SRTM (Shuttle Radar Topography Mission)** support erosion and watershed flow analysis (Gorelick et al., 2017). These tools ensure consistent global standards while empowering local stakeholders to implement evidence-based watershed management strategies (Roy et al., 2016).

6.4 Mobile tools for community-driven data collection

In developing countries, local community participation in data collection is critical for success. Mobile-based tools such as **KoboToolbox** and **Open Data Kit (ODK)** allow non-experts to collect environmental and socioeconomic data in standardized formats. These apps can capture GPS locations, photos, and other metadata automatically, thus ensuring consistency across datasets.

Community groups can use these mobile applications to record rainfall patterns, monitor soil conditions, or log observations on water quality. These tools are easy to use, even in areas with limited technical capacity, and their data can be integrated into larger regional or national databases for analysis (Tabbush, 2018). This approach not only standardizes data but also fosters local ownership and participation in watershed management (Tabbush, 2018).

6.5 Hydrological models

Standardized hydrological watershed models such as **SWAT (Soil and Water Assessment Tool)** and **WEPP (Water Erosion Prediction Project)** have made it easier to simulate and predict the impact of various land and water management interventions. These models require standardized inputs, such as soil characteristics, land use, land cover, DEM, and climate information, to ensure globally consistent and comparable outputs.

In sub-Saharan Africa, SWAT has been used successfully to predict the impact of agricultural practices on water quality and soil erosion. These models enable decision-makers to test different scenarios, such as reforestation or irrigation changes, before implementing them in the field (Flanagan et al., 2007).

6.6 Open data platforms

Global databases and open-access platforms provide ready-made datasets for regions with limited capacity for primary data collection. **FAO's AQUASTAT** provides standardized water statistics, while **HydroSHEDS** offers hydrological datasets such as river networks and watershed boundaries. These platforms simplify access to high-quality datasets while diminishing dependency on expensive local data collection efforts (Lehner et al., 2008).

Such platforms prove especially valuable in areas with underdeveloped local monitoring networks. By integrating these global datasets with local observations, watershed planners can create more comprehensive and reliable management plans (FAO, 2022).

6.7 Climate data integration

Climate change's impact on watersheds makes the integration of standardized climate data essential. Free resources such as **CHIRPS (Climate Hazards Group InfraRed Precipitation with Station Data)** and **CMIP6 (Coupled Model Intercomparison Project)** provide reliable historical and future climate projections. These datasets are invaluable for forecasting risks such as floods, droughts, and changes in seasonal water availability (Funk et al., 2015; Eyring et al., 2016).

Standardized formats ensure that these data can be directly used in watershed models or planning tools, thus helping decision-makers account for climate variability and build resilience into their strategies.

6.8 Citizen science programs

Citizen science initiatives, in which volunteers can collect data from local landscapes, help not only in gathering data but also in raising community awareness on watershed management issues. This approach can provide real-time information about a given watershed using various tools such as mobile apps.

6.9 Artificial intelligence and machine learning

Artificial intelligence and machine learning are revolutionizing watershed data collection through several innovative applications. By leveraging AI and ML in various methods, watershed researchers and managers can gather more accurate data. The following are some of the applications of AI and ML:

- ▶ AI algorithms can analyze historical water quality data to predict future conditions. For instance, researchers in the Chesapeake Bay watershed use ML models to forecast nutrient runoff based on weather patterns, land use, and agricultural practices. This helps in proactive management of water quality.
- ▶ AI can optimize the deployment and management of IoT sensors placed in watersheds. These sensors collect real-time data on streamflow, soil moisture content, and rainfall. In regions such as the Mississippi River Basin, ML algorithms can analyze these data to detect anomalies, such as unexpected spikes in turbidity, indicating pollution events.
- ▶ AI-driven image recognition tools can analyze drone or camera trap footage to monitor wildlife and habitats within watersheds. In the Sierra Nevada mountains, ML algorithms help track the presence and movement of species that are sensitive to changes in water availability and quality.
- ▶ AI chatbots and mobile applications can facilitate community reporting of pollution or habitat degradation. These data can then be aggregated and analyzed using ML to identify hotspots and trends across a watershed, thus empowering local stakeholders in environmental stewardship.
- ▶ Machine learning techniques can process large volumes of satellite imagery to assess land cover changes and monitor vegetation health. For example, NASA's Earth Observing System Data and Information System uses AI to detect changes in watershed surfaces, thus enabling rapid identification of deforestation or urbanization impacts.
- ▶ Machine learning can enhance flood forecasting by analyzing historical rainfall and river flow data alongside real-time meteorological data. For instance, the National Oceanic and Atmospheric Administration (NOAA) employs AI models to predict flooding in watersheds, thus allowing timely alerts and resource allocation.



7. Summary and Ways Forward

7.1 Analysis of biophysical problems

Brief and essential interpretations are needed for the data collected. The data then need to be analyzed and interpreted in relation to relevant watershed problems or management objectives. Data should be analyzed in relation to the most important biophysical problems in a watershed as described as follows.

Land use capability: Determining proper land use based on land capability or suitability is always the first step toward the protection and development of a watershed. A land use adjustment map can be produced by superimposing the current land use against land capability maps. This can help identify areas of serious overuse requiring urgent intervention to reverse degradation. It can pinpoint underused land with potential for increased economic output. The map provides the basis for appropriate use of watershed lands and those that require soil conservation measures.

Water resources and hydrology analysis: From the basic data collected, stream flow patterns can be analyzed, including annual/seasonal variations, maximum and minimum discharge rates, and water quality indicators such as turbidity and pollutants. Flood and drought patterns should be studied. Water use concerns, including consumption rates and quality, must be addressed. Findings can be compiled in an annual water resources and hydrology reports.

Hazards of erosion and sedimentation: The various sources and damages of erosion and sedimentation should be identified and analyzed, and potential hazards identified for successful conservation planning. The latter is very important since most watershed rehabilitation or protection is centered on the minimization of potential hazards. Special efforts should be made to analyze data collected on erosion and sedimentation. Data can be collected from field surveys, observations, interviews, soil loss and runoff plot data analysis, erosion modeling and prediction, storm frequency analysis, sediment delivery assessment, reservoir sedimentation data, and geological analysis for landslide risk. Treatment costs and potential benefits of control measures should be evaluated.

7.2 Maps, statistics, and interpretations

Maps and statistics are a major product of watershed data and watershed biophysical surveys. The kinds of maps and statistics produced depend on watersheds and management objectives. Key documentation should include maps showing baseline information (boundaries, sub-watersheds, villages, roads, etc.); topography (contours, elevations, landforms, streams, etc.); soil characteristics (types, depths, fertility, etc.); climate data (rainfall, temperature, evapotranspiration, etc.); a geology map (rock types, structures, displacement, morphology, etc.); slope classifications; current land use and coverage; land suitability and capability; land use adjustment recommendations; erosion and sediment source locations; hydro-

meteorological networks showing the location of climatic and stream gauging stations; and water resources (surface and underground). If needed, additional maps might include forest coverage, landslide risk areas, slope stability, and stream profiles.

7.3 Plans and management recommendations

Survey results and analyses should be periodically reported to watershed management bodies to guide strategic and annual operational planning. Based on survey data and analysis of thematic watershed data, the watershed management team should develop initial draft plans and management recommendations for further discussion.

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